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HEB collaborates with TCR signaling to upregulate *Id3* and enable $\gamma\delta$ T17 cell maturation in the fetal thymus

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eLife Assessment

The study provides **important** mechanistic insight into the transcriptional control of $\gamma\delta$ T17 development, elegantly demonstrating how HEB and *Id3* act sequentially and cooperatively to regulate $\gamma\delta$ T17 cell specification and maturation. The study provides **compelling** evidence that advances the understanding of E-Id protein dynamics in thymic T cell specification. The work is comprehensive, technically rigorous, and conceptually clear, and will be of interest to immunologists, developmental biologists, and those studying the molecular underpinnings of physiological outcomes.

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Abstract

T cells expressing the $\gamma\delta$ T cell receptor (TCR) develop in a stepwise process initiating at the $\alpha\beta/\gamma\delta$ T cell branchpoint followed by maturation and acquisition of effector functions, including the ability to produce interleukin-17 (IL-17) as $\gamma\delta$ T17 cells. Previous studies linked TCR signal strength and fate choices to the transcriptional regulator HEB (*Tcf12*) and its antagonist, *Id3*, but how these factors regulate different stages of $\gamma\delta$ T cell development has not been determined. We found that immature fetal $\gamma\delta$ TCR⁺ cells from conditional *Tcf12* knockout (HEB cKO) mice were defective in activating the $\gamma\delta$ T17 program at an early stage, whereas *Id3* deficient (*Id3*-KO) mice displayed a partial block in $\gamma\delta$ T17 maturation and a defect in IL-17 production. We also found that HEB cKO mice failed to upregulate *Id3* during $\gamma\delta$ T17 development, whereas HEB overexpression elevated the levels of *Id3* in collaboration with TCR signaling. Moreover, *Egr2* and HEB were bound to several of the same regulatory sites on the *Id3* gene locus in the context of early T cell development. Therefore, our findings reveal an interlinked sequence of events during which HEB and TCR signaling synergize to upregulate *Id3*, which enables maturation and acquisition of the $\gamma\delta$ T17 effector program.

One Sentence Summary

The transcription factor HEB synergizes with TCR signaling to upregulate *Id3*, which is required for the maturation of fetal IL-17-producing $\gamma\delta$ T cells.

Introduction

IL-17-producing $\gamma\delta$ T ($\gamma\delta$ T17) cells are critical effectors in immune responses to bacterial and fungal pathogens and contribute to tissue repair and barrier integrity, particularly at mucosal surfaces (Ribot, Lopes, & Silva-Santos, 2021 [↗](#)). Beyond host defense, $\gamma\delta$ T17 cells regulate adipose tissue homeostasis and metabolic function (Kohlgruber et al., 2018 [↗](#)). However, dysregulation of $\gamma\delta$ T17 cell function has been linked to autoimmunity and neuroinflammatory disorders, underscoring the importance of precisely controlled $\gamma\delta$ T17 cell development and function (Alves de Lima et al., 2020 [↗](#); Papotto, Reinhardt, Prinz, & Silva-Santos, 2018 [↗](#)).

While significant progress has been made in defining the stages of $\gamma\delta$ T cell development, the transcriptional networks that govern lineage commitment and effector fate specification remain incompletely understood. In mice, $\gamma\delta$ T17 cells arise exclusively during fetal and early neonatal thymic development. The earliest wave of $\gamma\delta$ T17 cells in the fetal thymus, which emerges around embryonic day 17 (E17), originates from T cell precursors expressing the V γ 6 V δ 1 T cell receptor (TCR) (W. Haas, Pereira, & Tonegawa, 1993 [↗](#); Shibata et al., 2008 [↗](#)). A second wave of V γ 4⁺V δ 5⁺ cells, which begins at E18, also gives rise to $\gamma\delta$ T17 cells (In et al., 2017 [↗](#); Sandrock et al., 2018 [↗](#)). By contrast, fetal-restricted V γ 5⁺V δ 1⁺ $\gamma\delta$ T cells are associated with the IFN- γ -producing $\gamma\delta$ T1 fate (Havran & Allison, 1990 [↗](#); Turchinovich & Hayday, 2011 [↗](#)). Most V γ 1⁺ cells, which appear just before birth and continue to develop in the adult thymus, also differentiate into $\gamma\delta$ T1 cells (Buus, Odum, Geisler, & Lauritsen, 2017 [↗](#)). Although V γ 4⁺ cells continue to be generated in the adult thymus, their capacity to give rise to innate $\gamma\delta$ T17 cells is significantly diminished (Chen et al., 2021 [↗](#); Sandrock et al., 2018 [↗](#); Yang et al., 2023 [↗](#)). Instead, adult V γ 4⁺ cells remain poised for peripheral polarization into either $\gamma\delta$ T17 or $\gamma\delta$ T1 fates (Schmolka et al., 2013 [↗](#)). Other subsets, such as IL-4 producing V γ 1⁺V δ 6.3⁺ cells, emerge primarily in neonatal and young mice (Grigoriadou, Boucontet, & Pereira, 2003 [↗](#)).

$\gamma\delta$ and $\alpha\beta$ T cells develop primarily from shared intrathymic T cell progenitors that lack expression of CD4 and CD8 (double-negative, DN) (Ciofani & Zuniga-Pflucker, 2010 [↗](#); Dudley, Girardi, Owen, & Hayday, 1995 [↗](#)). These precursors can be further subdivided by CD44 and CD25 expression, with the earliest thymic immigrants classified as CD44⁺CD25⁻, termed early thymic progenitors (ETPs). In the fetal thymus, DN2 (CD44⁺CD25⁺) cells can generate either $\gamma\delta$ T17 or $\gamma\delta$ T1 cells, while DN3 (CD44⁻CD25⁺) cells preferentially give rise to $\gamma\delta$ T1 cells or proceed to $\alpha\beta$ development following pre-TCR signaling in a process known as β -selection (Dutta, Zhao, & Love, 2021 [↗](#)). Successful progress through β -selection is followed by upregulation of CD8 and CD4, generating CD4⁺CD8⁺ double positive (DP) cells. Commitment to the $\gamma\delta$ lineage is initiated by $\gamma\delta$ TCR signaling, which induces lineage-specifying transcription factors such as *Sox13* (Gray et al., 2013 [↗](#); Malhotra et al., 2013 [↗](#)), followed by effector programming factors such as *Maf*, which are gradually upregulated as differentiation proceeds (Malhotra et al., 2013 [↗](#); Sagar et al., 2020 [↗](#)). These factors promote expression of lineage-defining cytokines such as IL-17 and their regulators, including ROR γ t (Zuberbuehler et al., 2019 [↗](#)).

TCR signal strength plays a central role in determining both the $\gamma\delta$ T cell lineage choice and $\gamma\delta$ T cell effector program (Lee, Stadanlick, Kappes, & Wiest, 2010 [↗](#); Zarin, Wong, Mohtashami, Wiest, & Zuniga-Pflucker, 2014 [↗](#)). Weak pre-TCR signals promote $\alpha\beta$ lineage progression, whereas intermediate $\gamma\delta$ TCR signaling favors $\gamma\delta$ T17 cell development, and stronger signals direct $\gamma\delta$ T1 differentiation (Haks et al., 2005 [↗](#); Zarin, Chen, In, Anderson, & Zuniga-Pflucker, 2015 [↗](#)). Signal strength is modulated by TCR affinity, proximal CD3 signaling, and cytokine crosstalk (Fahl, Coffey, et al., 2018; Fahl, Kappes, & Wiest, 2018 [↗](#); Michel et al., 2012 [↗](#); Muro et al., 2018 [↗](#)). Together, these factors converge on downstream signal transduction pathways, including the ERK–Egr–Id3 axis, which has been identified as a key mediator of TCR signal strength using manipulation of TCR ligands or TCR signaling pathways (Bain et al., 2001 [↗](#); Fahl, Kappes, et al., 2018; Lee et al., 2014 [↗](#); Munoz-Ruiz et al., 2016 [↗](#); Turchinovich & Hayday, 2011 [↗](#)).

Id3 encodes a dominant-negative HLH protein that inhibits the expression of E protein-dependent genes. *Id3* acts at the post-translational level by binding and sequestering E proteins, including HEB (encoded by *Tcf12*) and E2A (encoded by *Tcf3*), thereby preventing their DNA binding activity.

HEB and E2A orchestrate multiple stages of $\alpha\beta$ T cell development (D'Cruz, Lind, Wu, Fujimoto, & Goldrath, 2012; Jones & Zhuang, 2011; Jones-Mason et al., 2012; Leung et al., 2025). *Id3* expression is proportional to TCR signal strength, suggesting that it may influence T cell fate by titrating E protein activity (Lauritsen et al., 2009; Zarin et al., 2018). Moreover, *Id3* transcription is directly regulated by graded expression of Egr factors, linking TCR signal strength to E protein target genes (Lauritsen et al., 2009). We previously showed that HEB-deficient mice exhibit severe defects in $\gamma\delta$ T17 development, including impaired production of fetal $V\gamma 4^+$ $\gamma\delta$ T cells and dysregulated expression of key $\gamma\delta$ T17 regulators in the $V\gamma 6^+$ subset (In et al., 2017). HEB directly activates early $\gamma\delta$ T17 genes, such as *Sox4* and *Sox13*, which are repressed when *Id3* is overexpressed. However, the complete physiological role of HEB in $\gamma\delta$ T cell development remains unresolved.

Here, we investigated the roles of HEB and *Id3* *in vivo* by analyzing fetal $\gamma\delta$ T cell development in HEB^{f/f}, Vav-iCre (HEB conditional knockout, HEB cKO) and *Id3*-deficient (*Id3*-KO) mice using flow cytometry and single-cell RNA sequencing (scRNA-seq) of E18 thymocytes. Our results reveal a tiered disruption of $\gamma\delta$ T cell development in HEB cKO mice, including dysregulated TRVG and TRDV expression, lineage diversion to the $\alpha\beta$ program, and failure to activate key specification factors, including *Id3*. In contrast, $\gamma\delta$ T17 precursors in *Id3*-KO mice initiated the $\gamma\delta$ T17 specification program but failed to mature or produce IL-17. These findings suggest that HEB and *Id3* function in an interlinked negative feedback loop that reinforces $\gamma\delta$ lineage commitment and mediates the transition from specification to maturation during $\gamma\delta$ T17 cell development.

Results

Strategy for analyzing $\gamma\delta$ T cell developmental progression in the fetal thymus

To analyze $\gamma\delta$ T cell development by flow cytometry, we used a panel of antibodies that distinguished developmental stages and lineage fate choices in the mouse fetal thymus. $\gamma\delta$ T cells develop from DN2/3 cells in the thymus from precursors with both $\alpha\beta$ and $\gamma\delta$ T cell potential, with upregulation of CD8 (immature single positive, ISP) and CD4 marking commitment to the $\alpha\beta$ -T cell lineage (Figure 1A). $\gamma\delta$ Te (early $\gamma\delta$ T cell subsets prior to maturation; CD24⁺) cells arising from DN2/3 cells express $\gamma\delta$ TCR and CD24 on the cell surface but retain the ability to become either $\gamma\delta$ T cells or $\alpha\beta$ T cells (dotted arrow) (Coffey et al., 2014). Strong $\gamma\delta$ TCR signaling results in upregulation of CD73 (*Nt5e*) at the $\gamma\delta$ T1p stage (p = progenitor), followed by downregulation of CD24 (*Cd24a*) to yield mature CD27⁺CD24[−]CD73⁺CD44[−] $\gamma\delta$ T1 (IFN γ producing) cells. Signaling through $V\gamma 6^+$ or $V\gamma 4^+$ TCRs results in downregulation of CD24 and upregulate CD44 as $\gamma\delta$ T cell precursors differentiate into immature $\gamma\delta$ T17p cells, which mature into CD27[−]CD24[−]CD73[−]CD44⁺ $\gamma\delta$ T17 (IL-17-producing) cells (Sumaria, Grandjean, Silva-Santos, & Pennington, 2017). CD27 is expressed on all immature $\gamma\delta$ T cells and stays on in mature $\gamma\delta$ T1 cells but is downregulated during $\gamma\delta$ T17 maturation. Thus, detection of $\gamma\delta$ TCR, CD4, CD8, CD24, CD73, CD27, and V γ chains, and the genes that encode them, provides a solid framework for analyzing the impact of gene perturbations on fetal $\gamma\delta$ T cell development and maturation.

It is important to clarify the V γ and V δ chain nomenclature used in this study (Figure 1B). Cell surface proteins detected in flow cytometry assays were classified using the Tonegawa nomenclature (Heilig & Tonegawa, 1986). When referring to genes or gene transcripts, the International Immunogenetics Information System (IMGT) nomenclature (Lefranc, 2003) was used, except in scRNA-seq figures where R-Seurat program output gene names from the Mouse Genome Informatics (MGI) site were preserved (Baldarelli et al., 2024). The numbering is the same in all three systems except for V δ 1, which is equivalent to TRDV4 (IMTG) and Trdv4 (MGI).

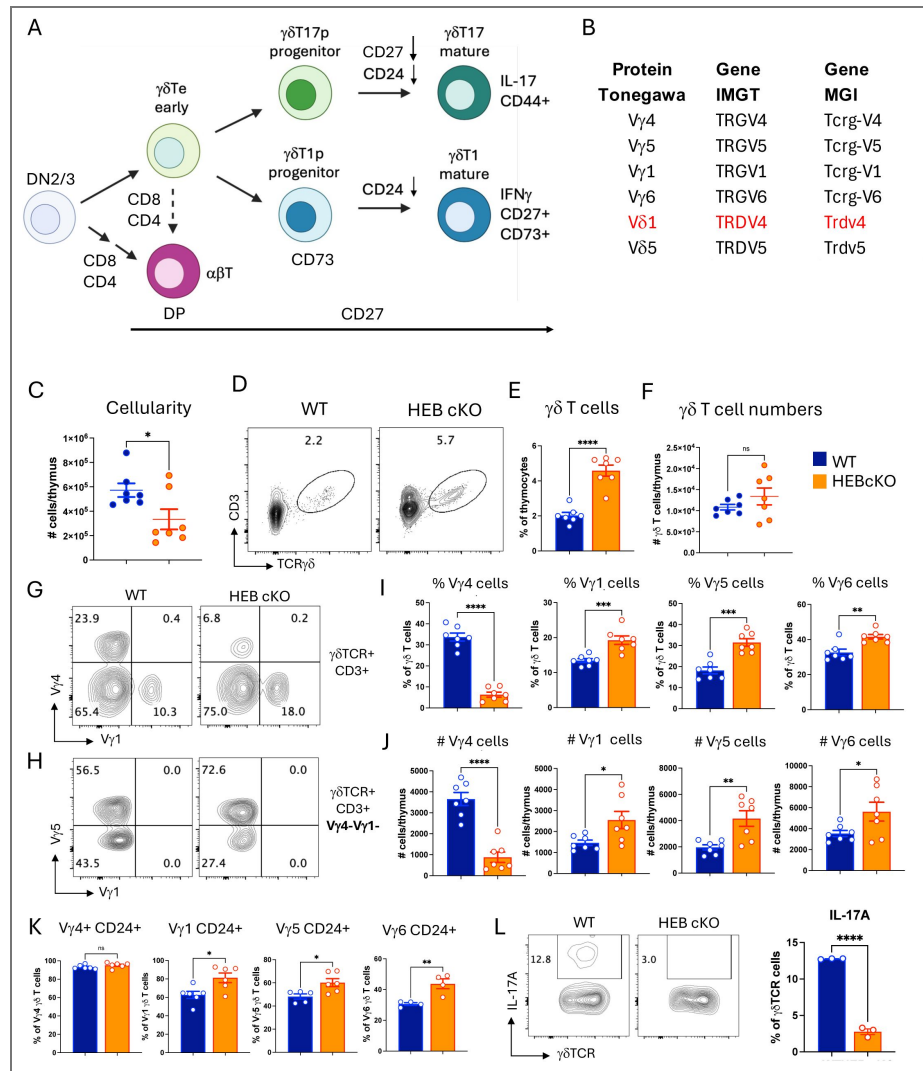


Figure 1. Partial block in early $\gamma\delta$ T cell development and decrease in V γ 4 cells in HEB-deficient mice.

A. Stages of fetal mouse $\gamma\delta$ T cell development. Thymocytes at the DN2 ($CD4^+CD8^+CD44^+CD25^+$) and DN3 ($CD4^+CD8^+CD44^+CD25^+$) stages of T cell development rearrange and express TCR γ , TCR δ , and TCR β genes. DN3 cells with productive TCR β chains expressing a pre-TCR are directed into the $\alpha\beta$ -T cell lineage, characterized by upregulation of CD4 and CD8. Cells that express a cell surface $\gamma\delta$ TCR but have not yet committed to the $\gamma\delta$ T cell lineage ($\gamma\delta$ Te, early $\gamma\delta$ T cells) can still be diverted into the $\alpha\beta$ T cell lineage (dotted line). $\gamma\delta$ TCR $^+$ cells receiving a strong signal become CD73 $^+$ $\gamma\delta$ T1 cell progenitors ($\gamma\delta$ T1p) that mature into IFN γ producing $\gamma\delta$ T1 cells, whereas cells that receive an intermediate signal become $\gamma\delta$ T17 cell progenitors ($\gamma\delta$ T17p) that mature into IL-17 producing $\gamma\delta$ T17 cells. Downregulation of CD24 and CD27, and upregulation of CD44, marks maturation of $\gamma\delta$ T17 cells, whereas $\gamma\delta$ T1 cell maturation is characterized by downregulation of CD24 and maintenance of CD73 and CD27 expression.

B. $\gamma\delta$ T cell nomenclature. V γ and V δ chain genes and proteins can be identified by several different naming systems. Here we use the Tonegawa nomenclature to refer to the proteins, and the International Immunogenetics Information System (IMGT) for the genes and transcripts. R-Seurat-generated plots use the Mouse Genome Informatics (MGI) nomenclature. The numbering for genes and proteins in these three systems are identical except for the TRDV4 gene, which encodes the V γ 1 protein (highlighted in red).

C. Absolute numbers of cells per thymus in WT (blue) and HEB cKO (orange) E18 fetal mice. D, E. Percentages of $\gamma\delta$ T cells in WT and HEB cKO fetal thymus. F. Absolute number of $\gamma\delta$ T cells per thymus in WT and HEB cKO fetal thymus. G. Flow cytometry plots of V γ 4 $^+$ and V γ 1 $^+$ cells within the $\gamma\delta$ T cell population in WT and HEB cKO fetal thymus. H. Flow cytometry plot of V γ 5 and V γ 6 (V γ 1 $^+$ V γ 5 $^+$) expression on cells within the V γ 1 $^+$ V γ 4 $^+$ population. I. Percentages of V γ 1 $^+$, V γ 4 $^+$, V γ 5 $^+$, and V γ 6 $^+$ cells out of all $\gamma\delta$ T cells in the WT and HEB cKO fetal thymus. J. Absolute numbers of V γ 1 $^+$, V γ 4 $^+$, V γ 5 $^+$, and V γ 6 $^+$ cells per WT and HEB cKO fetal thymus. K. Percentages of immature (CD24 $^+$) and mature (CD24 $^-$) $\gamma\delta$ T cells out of all $\gamma\delta$ T cells in each V γ subset in WT and HEB cKO fetal thymus. L. Expression of IL-17A protein in $\gamma\delta$ T cells from E18 thymus stimulated with PMA/ionomycin as assessed by intracellular staining. Blue = WT, orange = HEB cKO. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

HEB deficiency impairs Vy4 cell development and inhibits functional maturation of $\gamma\delta$ T17 cells in the E18 fetal thymus

To investigate how HEB loss affects fetal $\gamma\delta$ T cell development, we crossed HEB^{f/f} mice with HEB^{f/f} Vav-iCre mice to generate littermates without Cre (wildtype, WT) or with Cre (HEB cKO). HEB cKO mice lack HEB in all hematopoietic cells, as previously described (In et al., 2017; Welinder et al., 2011). At embryonic day 18 (E18), HEB cKO thymocytes showed reduced cellularity (Figure 1C) and a developmental block at the ISP-to-DP transition (Figure 1 - figure supplement 1), as previously reported in adult HEB-deficient mice (R. Barndt, Dai, & Zhuang, 1999; Wojciechowski, Lai, Kondo, & Zhuang, 2007). Although $\gamma\delta$ T cells were proportionally increased in HEB cKO mice (Figure 1D, E), their absolute numbers were comparable to WT mice (Figure 1F), indicating that the reduced cellularity was primarily due to the loss of DP thymocytes.

We next analyzed Vy chain subsets. Vy6 cells were identified using an exclusion strategy, as cells lacking expression of Vy1, Vy4, and Vy5 (Figure 1G, H). It should be noted that the flow cytometry plots show the percentage of Vy5⁺ and Vy5⁻ (Vy6) cells out of the Vy1 Vy4⁺ population (Figure 1H), whereas the bar graph shows the percentage of Vy6 cells out of all $\gamma\delta$ TCR⁺ cells (Figure 1I). HEB cKO mice exhibited a marked reduction in the frequency (Figure 1G, I) and absolute numbers (Figure 1J) of Vy4⁺ cells, and a corresponding increase in the other Vy subsets (Figure 1G, H, I, J). Immature (CD24⁺) cells were more prevalent across all Vy subsets in HEB cKO mice, except for Vy4⁺ cells, which were primarily CD24⁺ and thus had not yet matured, in either WT or HEB cKO mice (Figure 1K). Furthermore, fetal HEB cKO $\gamma\delta$ T cells showed a major impairment in IL-17 production in response to PMA/ionomycin stimulation (Figure 1L), consistent with our prior fetal thymic organ culture (FTOC) results (In et al., 2017). These findings indicate that even in the intact E18 thymic environment, HEB is essential for $\gamma\delta$ T17 cell development.

Single-cell transcriptomic analysis reveals a role for HEB in establishing $\gamma\delta$ T cell identity

To assess the transcriptomic impact of HEB deficiency, we performed single-cell RNA sequencing (scRNA-seq) on $\gamma\delta$ TCR⁺ thymocytes sorted from E18 WT and HEB cKO littermates. After quality control, we merged the datasets, excluded myeloid cells, and regressed out cell cycle genes using Seurat (Y. Hao et al., 2021). Our analysis identified eight distinct clusters (numbered 0 to 7) visualized as a UMAP (Figure 2A). WT cells were enriched in clusters 2 and 4, while HEB cKO cells dominated clusters 0, 1, and 3, whereas Clusters 5–7 showed a more balanced distribution (Figure 2B, C). To annotate the clusters, we curated 90 $\gamma\delta$ T cell subset-defining genes from prior studies (Inacio et al., 2025; Liu et al., 2020; Mistri et al., 2024; Narayan et al., 2012; Sagar et al., 2020; Spidale et al., 2018; Yang et al., 2023) (Figure 2 – figure supplement 1). The top 10 differentially expressed genes were used to generate a clustered dot plot (Figure 2D), which we used to identify each cluster. We classified clusters 2 and 6 as early $\gamma\delta$ T cells (randomly assigned as $\gamma\delta$ Te1 and $\gamma\delta$ Te2), clusters 1 and 4 as $\gamma\delta$ T17 progenitors ($\gamma\delta$ T17p), cluster 0 as $\gamma\delta$ T1 progenitors ($\gamma\delta$ T1p), cluster 7 as mature $\gamma\delta$ T1 cells ($\gamma\delta$ T1), cluster 5 as mature $\gamma\delta$ T17 cells ($\gamma\delta$ T17), and cluster 3 as $\alpha\beta$ lineage-like cells ($\alpha\beta$ T).

Quantification of WT and HEB cKO cells per cluster (Figure 2E) revealed that the $\gamma\delta$ T17p clusters segregated by genotype into WT ($\gamma\delta$ T17pw) and HEB cKO ($\gamma\delta$ T17pk) cells. HEB cKO cells were also over-represented in the $\alpha\beta$ T cluster and under-represented in the $\gamma\delta$ Te1 cluster. An unbiased heatmap of the top 10 most differentially expressed genes across all clusters further validated our assigned identities, and revealed additional genes associated with these subsets (Figure 2F). These findings suggest that the loss of HEB impairs the early $\gamma\delta$ T cell program and allows diversion toward the $\alpha\beta$ T cell lineage.

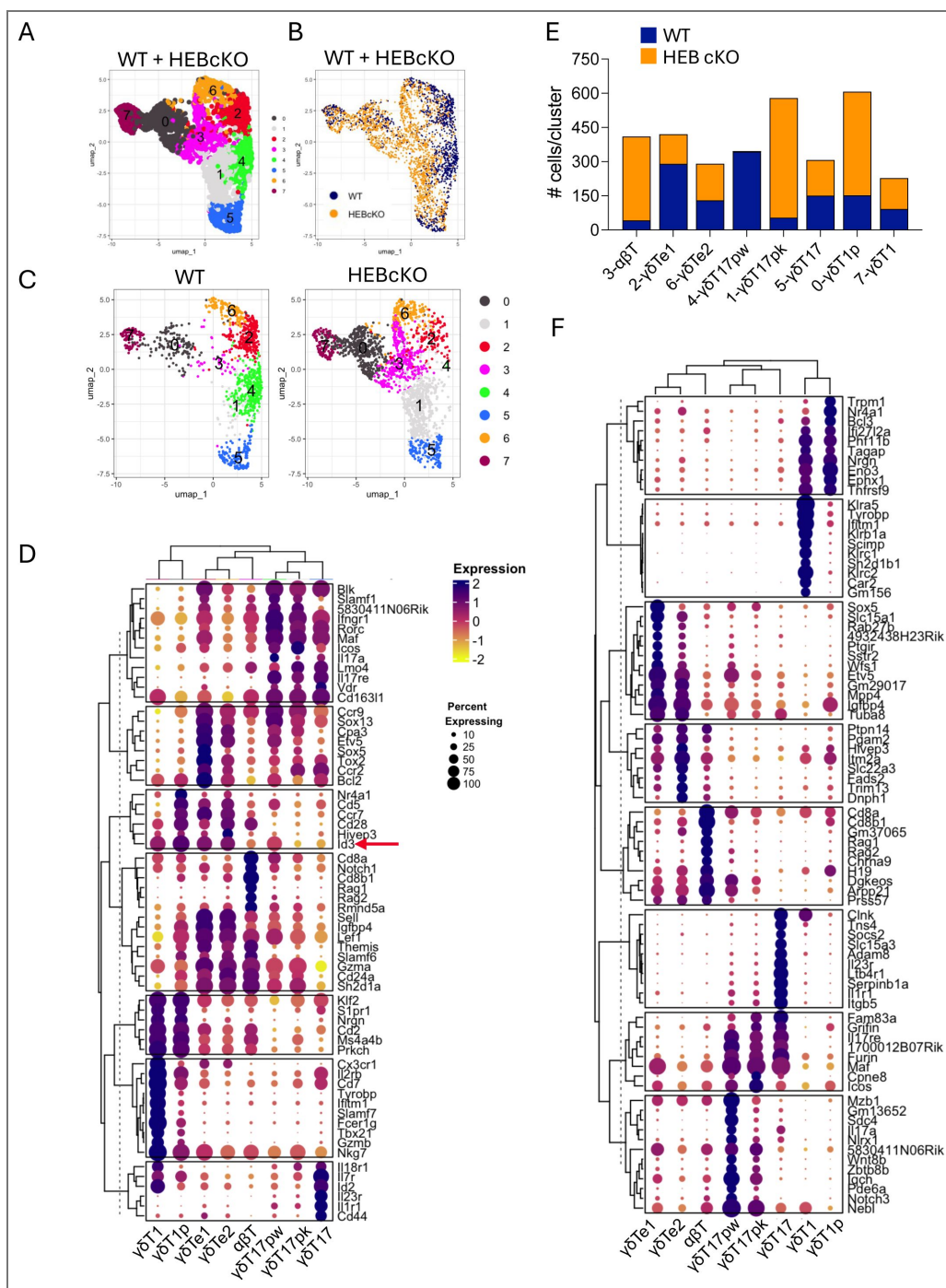


Figure 2. Identification of $\gamma\delta$ T cell subsets in WT and HEB cKO fetal thymus by scRNA-seq.

$\gamma\delta$ T cells were sorted from E18 fetal thymuses from WT and HEB cKO mice, pooled according to genotype, and subjected to scRNA-sequencing and analysis. **A**. UMAP plots depicting merged WT and HEB cKO cells in 8 clusters (0-7). **B**. Grouped UMAP showing the distribution of WT (blue) and HEB cKO (orange) cells across all clusters. **C**. Split UMAP plots showing the distribution of cells in WT (left) and HEB cKO (right) clusters; note that cluster 4 is restricted to WT cells, and cluster 1 is heavily biased towards HEB cKO cells. **D**. Genes previously identified as signatures for developmental and functional $\gamma\delta$ T cell subsets were compiled from previously published reports. The top ten most differentially expressed genes from this list were visualized as a clustered dot plot which was used to assign cluster identities. Two clusters corresponding to early $\gamma\delta$ T cells were randomly named $\gamma\delta$ Te1 and $\gamma\delta$ Te2. **E**. Numbers of WT (blue) and HEB cKO (orange) cells per cluster. **F**. Unbiased clustered dot plot of the top ten most differentially expressed genes across all clusters. In the clustered dot plots, the percentage of cells expressing the gene in each cluster is depicted by the size of the dot, and the color indicates the relative magnitude of expression across clusters.

HEB deficiency suppresses TRDV5 expression and promotes TRDV4 expression

We next confirmed that WT and HEB cKO cells within each cluster represented equivalent developmental subsets (Figure 2 - figure supplement 2). All immature $\gamma\delta$ T cell clusters expressed *Cd24a* and *Cd27*, while mature subsets lacked *Cd24a* (Figure 2 - figure supplement 2A). Among mature cells, only $\gamma\delta$ T1 cells expressed *Cd27* and *Nt5e* (CD73), whereas $\gamma\delta$ T17 cells were uniquely *Cd44*-positive, consistent with our assignments. We also analyzed TRGV and TRDV gene expression (Figure 2 - figure supplement 2B, C). In WT cells, TRGV4 and TRDV5 were co-expressed in $\gamma\delta$ Te and $\gamma\delta$ T17p populations but were greatly diminished in HEB cKO cells. TRGV6 and TRGV5 transcripts were detectable in $\gamma\delta$ Te subsets from HEB cKO mice, consistent with a delay in $V\gamma 6^+$ and $V\gamma 5^+$ $\gamma\delta$ T cell maturation, whereas TRDV4 was expressed more broadly and at higher levels in all immature HEB cKO $\gamma\delta$ T cell subsets relative to WT counterparts. These results indicate that HEB plays an important role in maintaining the subset-specific and magnitude of TRGV5 and TRDV4 expression during $\gamma\delta$ T cell development.

HEB deficiency impairs early $\gamma\delta$ T cell signatures and enhances $\alpha\beta$ -T lineage features

To further define differences between WT and HEB cKO cells at the transcriptomic level, we constructed gene modules composed of suites of signature genes for the $\gamma\delta$ Te1, $\gamma\delta$ Te2, $\alpha\beta$ T, $\gamma\delta$ T17p, $\gamma\delta$ T17, $\gamma\delta$ T1p, and $\gamma\delta$ T1 subsets, derived from the analysis of merged WT and HEB cKO cells (Figure 2D, F [↗](#); see Methods). These modules were used to assign scores in WT (left) and HEB cKO (right) cells, which were visualized using split dot plots.

This analysis revealed a pronounced loss of the $\gamma\delta$ Te1 gene signature in HEB cKO cells (Figure 3A [↗](#)), alongside a notable increase in the $\alpha\beta$ T signature (Figure 3B [↗](#)). The $\gamma\delta$ Te2, $\gamma\delta$ T17p, and $\gamma\delta$ T17 gene signatures were also diminished in HEB cKO cells relative to WT, whereas the $\gamma\delta$ T1p signature was enriched. We also examined expression of individual genes that were diagnostic for $\gamma\delta$ T cell subsets (Figure 3C [↗](#)) versus committed $\alpha\beta$ T cells (Figure 3D [↗](#)). In the $\gamma\delta$ Te1 and $\gamma\delta$ Te2 subsets of HEB cKO mice, *Sox13* and *Etv5* were reduced and *Cd8b1* and *Dgkeos* were elevated relative to their WT counterparts. In $\gamma\delta$ T17p cells, $\alpha\beta$ T-lineage genes were not detectable, but *Sox13* and *Etv5* were still lower in HEB cKO cells than WT cells, potentially decoupling the specification and commitment events. A less dramatic reduction in *Il1r1* was observed in HEB cKO $\gamma\delta$ T17 cells (Figure 3E [↗](#)), and the levels of *Nrgn* and *Eomes* were similar in $\gamma\delta$ T1 lineage cells (Figure 3F [↗](#)). These results suggest that the impact of HEB deficiency at the transcriptomic level was more pronounced in early $\gamma\delta$ T cell subsets than in mature $\gamma\delta$ T cells.

HEB deficiency obstructs $\gamma\delta$ T17 progenitor development and dampens TCR signal strength

The $\gamma\delta$ T17p subset segregated into distinct clusters in WT and HEB cKO cells, highlighting a critical stage of HEB-dependent regulation. To explore this further, we performed an unbiased differential gene expression analysis in WT cells versus HEB cKO cells and visualized the results using an enhanced volcano plot with stringent thresholds ($\log_2FC > 0.5$, $-\log_{10}P > 25$) (Figure 4A [↗](#)). Many of the top differentially expressed genes were signature markers of $\gamma\delta$ T17 differentiation, including *Blk*, *Sox13*, and *Etv5*, all of which were downregulated in HEB cKO cells. *Trdv4* was among the most upregulated genes in HEB cKO $\gamma\delta$ T17p cells.

We next generated a list of differentially expressed genes between WT and HEB cKO $\gamma\delta$ T17p cells using less stringent criteria (adj P value > 0.001 , $\log_2Fc > 0.25$) (Figure 4 - figure supplement 1). This list was subjected to gene ontology analysis using ShinyGO, which resulted in a list of KEGG pathways with reduced gene representation in HEB cKO $\gamma\delta$ T17p cells. We observed significant depletion of genes pathways related to TCR signaling (T cell receptor signaling, PD-1 checkpoint, NF- κ B signaling) and Th1, Th2, and Th17 differentiation in HEB cKO cells relative to WT cells (Figure 4 - figure supplement 2). Three key markers of TCR signal strength, *Cd5*, *Cd69*, and *Egr1*,

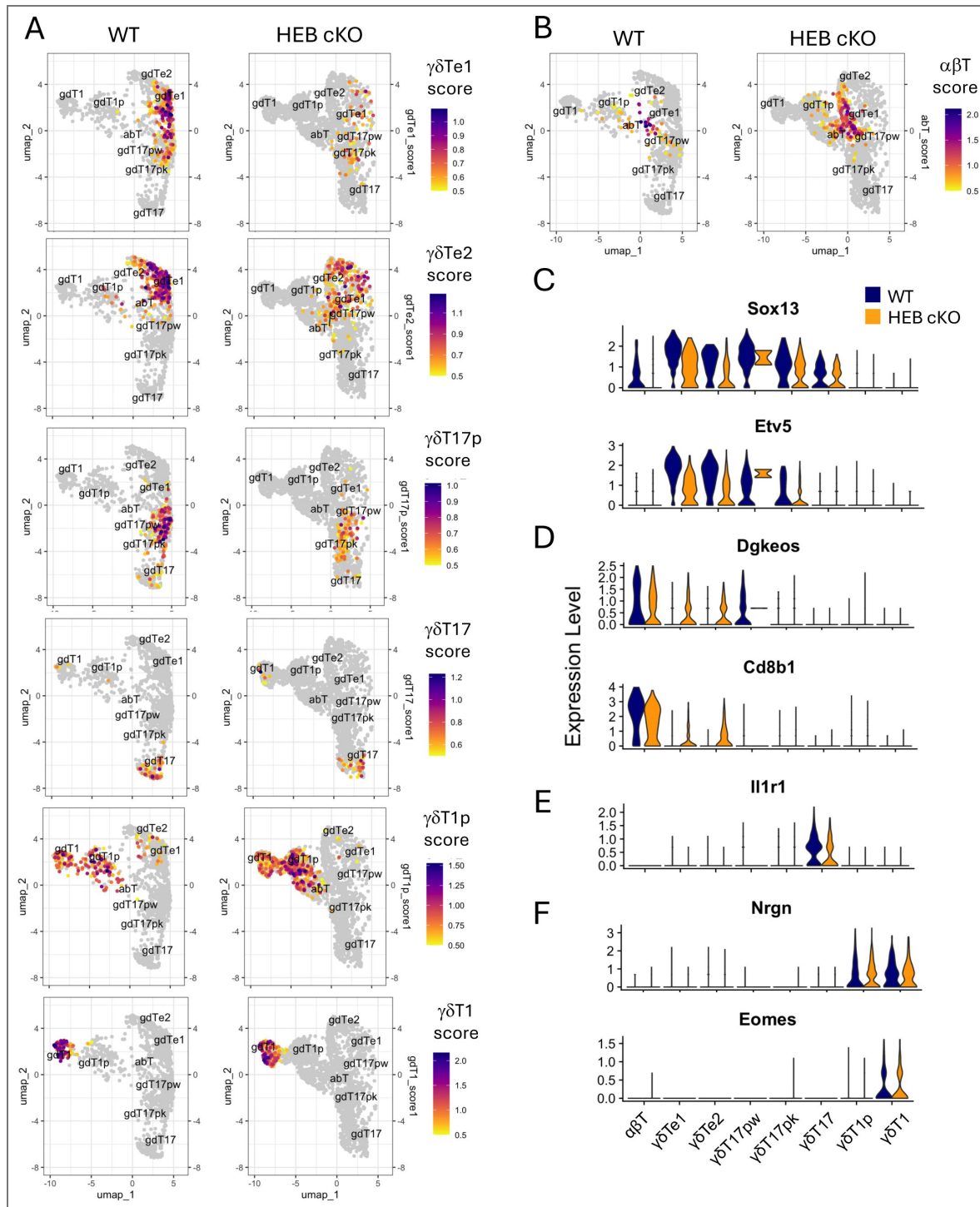


Figure 3. The $\alpha\beta$ T gene program is expanded and the $\gamma\delta\text{T17}$ precursor gene program is lost in HEB cKO cells.

$\gamma\delta$ T cells were sorted from E18 fetal thymuses from WT and HEB cKO mice were pooled according to genotype and subjected to scRNA-sequencing and analysis. A, B. Gene modules were generated from subset-biased genes, and cells were scored for each module. Module scores are depicted as split feature plots, with WT plots on the left and HEB cKO plots on the right. Module scores that characterize $\gamma\delta$ T cell subsets are shown in (A), and a module score for the $\alpha\beta$ -T lineage is shown in (B). C–G. Split violin plots of genes that typify different $\gamma\delta$ T cell subsets as follows: C) $\gamma\delta\text{Te}/\gamma\delta\text{T17p}$ cells, (D) $\alpha\beta$ T cells, (E) $\gamma\delta\text{T17}$ cells, (F) $\gamma\delta\text{T1p}/\gamma\delta\text{T1}$ cells, and (G) $\gamma\delta\text{T1}$ cells. Blue = WT, orange = HEB cKO.

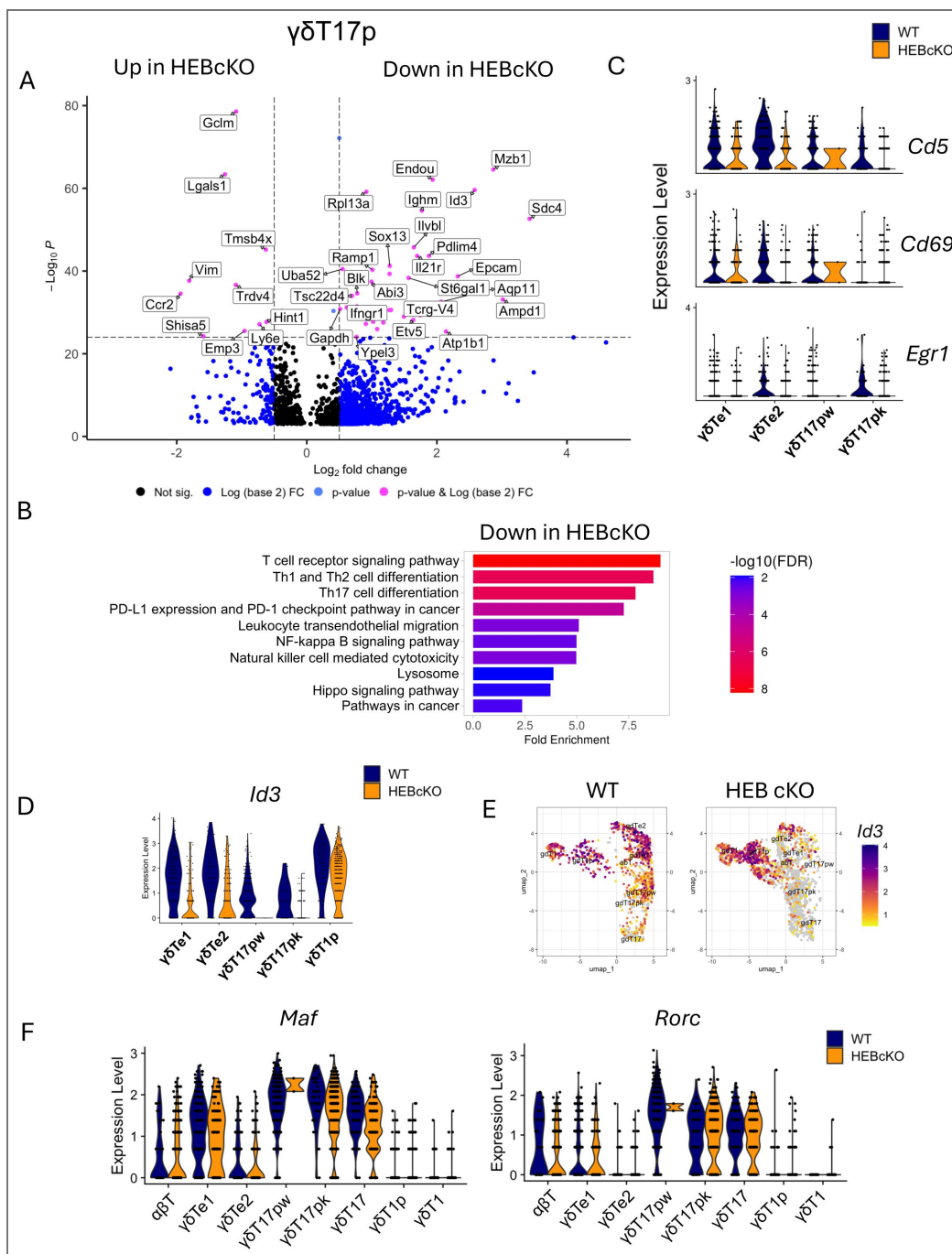


Figure 4. Decreases in T cell effector differentiation and TCR signaling genes in $\gamma\delta T17p$ cells from HEB cKO mice.

A. Volcano plots showing differential gene expression in $\gamma\delta T17p$ cells from WT versus HEB cKO fetal thymus. Genes expressed at higher levels in HEB cKO cells are on the left, and genes expressed at lower levels in HEB cKO cells are on the right. Significance (pink) was set at $\text{Log}_2\text{FC} > 0.5$ and $\text{Log}_{10}P < 10^{-25}$. B. Gene ontology analysis of genes significantly reduced in HEB cKO $\gamma\delta T17p$ cells relative to WT, with significance set at avg $\text{Log}_2\text{FC} > 0.25$ and adjusted P value < 0.001 . Bar plots show pathway enrichment (Fold enrichment) and significance by false discovery rate (FDR) for each functional category defined in the KEGG pathway list. Minimum genes for pathway inclusion was set at 5 and FDR cutoff was set at 0.05. C. Relative expression of genes associated with strong TCR signaling in WT and HEB cKO cells in each cluster. D. Relative expression of *Id3* in immature $\gamma\delta$ T cell subsets from WT and HEB cKO mice. E. Split feature plots showing expression of *Id3* across all clusters in WT versus HEB cKO cells. F. Relative expression of *Maf* and *Rorc* in WT versus HEB cKO $\gamma\delta$ T cell subsets. WT = blue, HEB cKO = orange.

were reduced in both $\gamma\delta$ Te2 and $\gamma\delta$ T17p subsets in HEB cKO mice (Figure 4C). *Id3* expression was markedly decreased in HEB cKO $\gamma\delta$ Te cells and nearly absent in $\gamma\delta$ T17p cells but remained intact in $\gamma\delta$ T1p and $\gamma\delta$ T1 cells (Figure 4D, E). We also examined the expression *Maf* and *Rorc*, two key regulators of $\gamma\delta$ T17 maturation, and found that they were expressed similarly between HEB cKO and WT cells in each cluster (Figure 4F). Therefore, HEB modulates TCR signaling during $\gamma\delta$ T17 specification but appears to be dispensable for the expression of $\gamma\delta$ T17 cell maturation factors.

***Id3* expression in $\gamma\delta$ T17 progenitors is controlled through HEB-dependent mechanisms**

To further elucidate the landscape of E protein and Id protein expression during $\gamma\delta$ T cell development, we examined the expression of *Tcf12* (encodes HEB), *Tcf3* (encodes E2A), *Id3*, and *Id2* in each cluster (Figure 4 – figure supplement 3). *Id1* and *Id4* transcripts were not detectable in any of our datasets. It should be noted that the *Tcf12* deletion occurs in one of the last exons of a 200 kb gene locus, and although the protein is absent (R. Barndt et al., 1999), some mRNA expression can still be detected. In WT mice, $\gamma\delta$ Te cells co-expressed *Tcf12*, *Tcf3*, and *Id3* (Figure 4 - figure supplement 3A-C, E), consistent with dynamic regulation of E protein-dependent target genes expression during the $\alpha\beta/\gamma\delta$ fate choice (Murre, 2019). $\gamma\delta$ T17p cells co-expressed *Tcf12* and *Tcf3*, whereas $\gamma\delta$ T1 cells co-expressed *Tcf3* and *Id3* (Figure 4 - figure supplement 3B, D). *Id2* expression was restricted to mature $\gamma\delta$ T17 and mature $\gamma\delta$ T1 cells in both WT and HEB cKO mice. *Tcf3* expression was also unaffected by a lack of HEB, indicating that *Tcf3* and *Id2* expression are HEB-independent (Figure 4 - figure supplement 3E, F). *Id3* expression was disrupted in HEB cKO cells in $\gamma\delta$ Te and $\gamma\delta$ T17 cells but not in $\gamma\delta$ T1p cells (Figure 4 - figure supplement 3C, D). This analysis reveals that *Id3* expression was disrupted only in cells that normally express *Tcf12*.

Loss of *Id3* impairs CD73 upregulation and $\gamma\delta$ T17 cell function in the fetal thymus

Given the clear dependence of *Id3* expression on HEB in specific $\gamma\delta$ T cell subsets, we next investigated how loss of *Id3* itself affects $\gamma\delta$ T cell development using E18 fetal thymocytes from *Id3* knockout (*Id3*-KO) mice. Total thymic cellularity in *Id3*-KO mice was comparable to WT controls (Figure 5A), but the proportion of $\gamma\delta$ T cells among total thymocytes was reduced (Figure 5B, C). Analysis of V γ chain usage revealed no major differences between WT and *Id3*-KO mice, although there was a slight increase in the proportion of V γ 6⁺ cells in *Id3*-KO $\gamma\delta$ T cells (Figure 5E, F). Interestingly, very few *Id3*-KO $\gamma\delta$ T cells expressed CD73 (encoded by *Nt5e*) (Figure 5G, H), including V γ 5⁺ and V γ 1⁺ cells (Figure 5 - figure supplement 1). *Nt5e* expression is upregulated in response to strong TCR signaling during $\gamma\delta$ T cell development, and it can also be induced in response to short-term stimulation (Buus, Geisler, & Lauritsen, 2016; Coffey et al., 2014). To determine whether the deficiency in CD73 expression in *Id3*-KO mice reflected an expansion of CD73[−] cells or a failure to induce *Nt5e*, we cultured E18 fetal thymocytes from WT and *Id3*-KO mice with PMA/ionomycin (P/I) for four hours and measured CD73 expression in $\gamma\delta$ T cells by flow cytometry (Figure 5I, J). While CD73 was robustly induced in a substantial fraction of CD27⁺ $\gamma\delta$ T cells from WT mice, it remained nearly undetectable in *Id3*-KO $\gamma\delta$ T cells, indicating a direct role for *Id3* in the regulation of *Nt5e* expression.

Mature CD73[−] $\gamma\delta$ T cells typically exhibit a bias toward the $\gamma\delta$ T17 lineage (In et al., 2017). We hypothesized that loss of *Id3* might re-direct $\gamma\delta$ T1-fated cells into the $\gamma\delta$ T17 lineage. However, upon stimulation, *Id3*-KO $\gamma\delta$ T cells failed to produce IL-17 (Figure 5K, L), indicating that *Id3* deficiency does not promote $\gamma\delta$ T17 differentiation. These findings highlight a critical role for *Id3* in the functional maturation of both CD73⁺ and CD73[−] $\gamma\delta$ T cells.

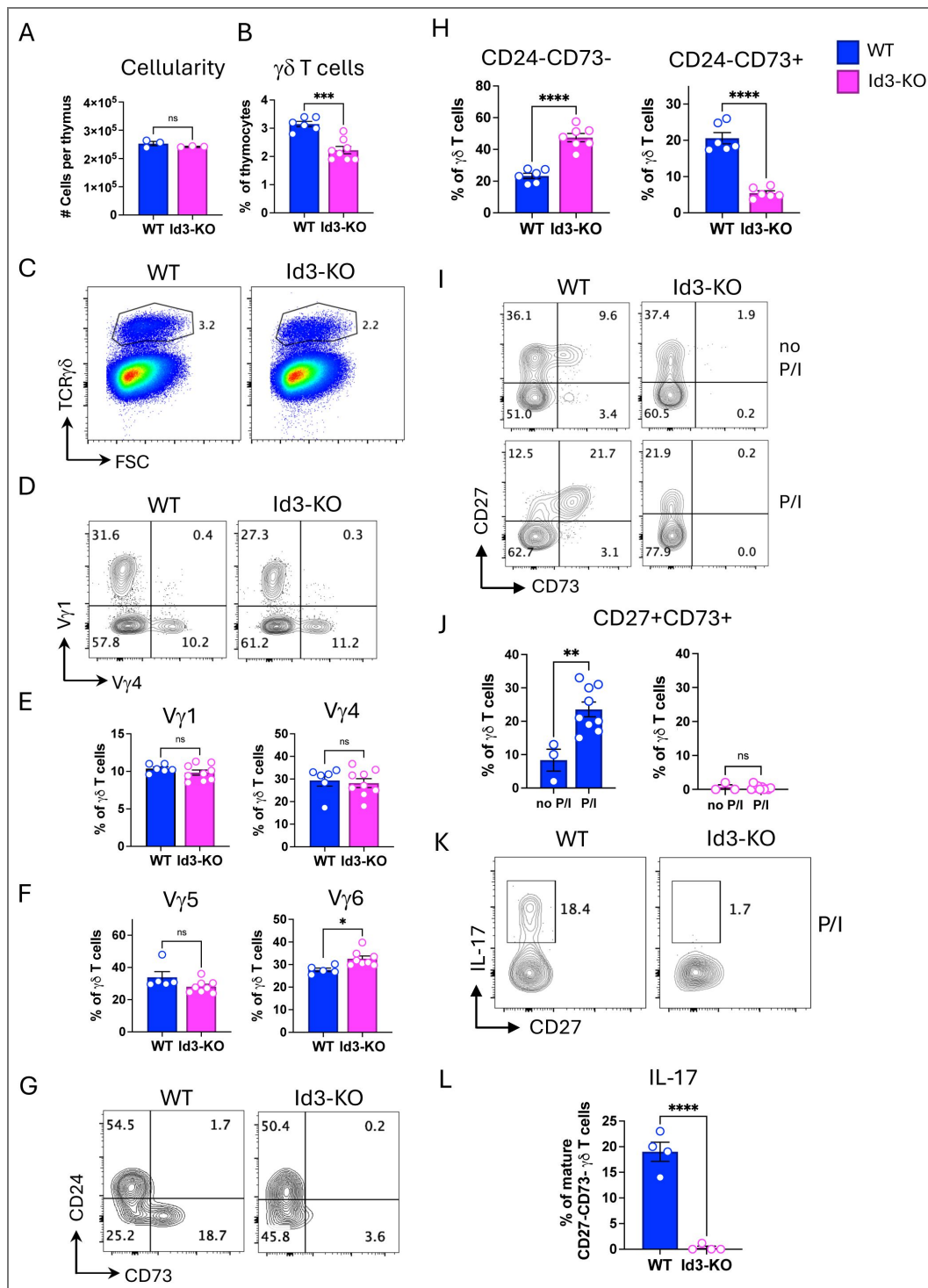


Figure 5. Fetal $\gamma\delta$ T cells from Id3-KO mice are defective in CD73 upregulation and IL-17 production.

A. Absolute numbers of cells per E18 fetal thymus from WT and Id3-KO littermate mice. B, C. Quantification (B) and flow cytometry plots (C) of the percentages of $\gamma\delta$ T cells out of all thymocytes. D, E. Flow cytometry plots (D) and quantification (E) of V γ 1⁺ and V γ 4⁺ out of all $\gamma\delta$ T cells. F. Percentages of V γ 5⁺ and V γ 6⁺ out of all $\gamma\delta$ T cells. G. Flow cytometry plots of CD24 and CD73 expression in $\gamma\delta$ TCR⁺ cells. H. Quantification of mature (CD24⁺) CD73⁺ and CD73⁻ $\gamma\delta$ T cells out of all $\gamma\delta$ T cells. I. Flow cytometry plots of expression of CD27 and CD73 expression in unstimulated (top) and stimulated (bottom) $\gamma\delta$ T cells. J. Percentages of CD27⁺CD73⁺ cells out of all $\gamma\delta$ T cells under unstimulated or stimulated conditions. K, L. Flow cytometry (K) and quantification (L) of the percentages of CD27⁺CD73⁻CD24⁺ (primarily mature V γ 6) cells expressing IL-17 in response to stimulation. Blue = WT, pink = Id3-KO. P/I = phorbol 12-myristate 13-acetate (PMA) + ionomycin. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Loss of Id3 disrupts expression of $\gamma\delta$ T17 cell maturation transcription factors

To further investigate how Id3 deficiency affects $\gamma\delta$ T cell development, we measured intracellular expression of PLZF (encoded by *Zbtb16*) and MAF in E18 $\gamma\delta$ T cells from WT and Id3-KO mice by flow cytometry (Figure 6C). PLZF is expressed in innate fetal/neonatal $\gamma\delta$ T cells and adult iNKT and IL-4-producing $\gamma\delta$ T cells (Alonzo et al., 2010; Kreslavsky et al., 2009; Lu, Cao, Zhang, & Kovalovsky, 2015). To compare developmental stages within V γ subsets between WT and Id3-KO $\gamma\delta$ T cells, we gated on: 1) immature (CD24⁺) V γ 4 cells, which comprised the majority of V γ 4 cells, 2) immature (CD24⁺) V γ 6 cells, and 3) mature (CD24⁻) V γ 6 cells (Figure 6A). This analysis yielded three populations, based on PLZF and MAF expression: PLZF⁺MAF⁺, PLZF⁺MAF⁻, and PLZF⁻MAF⁻ cells (Figure 6B).

In WT mice, most immature V γ 4 and V γ 6 cells expressed PLZF, with about half of the PLZF⁺ cells co-expressing MAF. Nearly all mature V γ 6 cells co-expressed PLZF and MAF. Immature V γ 4 and V γ 6 subsets in Id3-KO mice had reduced proportions of PLZF⁺MAF⁺ cell proportions and increased percentages of PLZF⁻MAF⁻ cells relative to WT. Mature V γ 6 cells in Id3-KO mice exhibited a more severe disruption, with only ~50% co-expressing PLZF and MAF, and significant increases within the PLZF⁺MAF⁻ or PLZF⁻MAF⁻ subsets. Mean fluorescence intensity analysis revealed substantially lower PLZF protein levels in Id3-KO cells across all subsets, especially immature cells, suggesting that this defect is not solely due to delayed maturation (Figure 6C).

Id3 deficiency promotes the $\alpha\beta$ -T lineage but does not disrupt the expression of early $\gamma\delta$ T cell regulators

To investigate population dynamics and gene expression changes in Id3-KO fetal thymocytes, we performed scRNA-seq on CD4⁺CD8⁺ E18 fetal thymocytes. This strategy was designed to capture $\gamma\delta$ T-biased cells with low surface $\gamma\delta$ TCR that might be missed by flow cytometric cell sorting. After quality control, WT and Id3-KO datasets were merged and subjected to dimensionality reduction, which identified eleven clusters (0–10) (Figure 6 – figure supplement 1A). Violin plots of lineage and subset-specific genes identified two $\gamma\delta$ T cell clusters (Figure 6 – figure supplement 1B), which were computationally isolated and reclustered. This process yielded four new $\gamma\delta$ T cell clusters (0–3), which we identified as $\gamma\delta$ Te, $\gamma\delta$ T17p, $\gamma\delta$ T17, and $\gamma\delta$ T1 cells (Figure 6D, E), using the same strategy shown in Figure 2 (Figure 6G). This scRNA-seq dataset lacked a $\gamma\delta/\alpha\beta$ T lineage cluster, likely due to low cell numbers and/or exclusion of CD4⁺ and CD8⁺ cells in the enrichment strategy. However, flow cytometry of E18 fetal thymocytes revealed a significantly higher proportion of TCR $\gamma\delta$ ⁺ cells co-expressing CD4 and CD8 in Id3-KO mice than in WT mice, indicating a bias toward the $\alpha\beta$ T cell program (Figure 6 – figure supplement 2). Notably, our scRNA-seq analysis showed that $\gamma\delta$ T cell specification genes (*Sox13*, *Etv5*) and *Tcf12* were unaffected by the decrease in *Id3* (Figure 6H). These results decouple the loss of *Id3* from the disruption of early $\gamma\delta$ T cell regulators in HEB cKO mice.

Id3 is required for the induction of $\gamma\delta$ T17 regulators at the transcriptional level

Compared to WT, the Id3-KO mice had increased numbers of $\gamma\delta$ T17p cells and fewer mature $\gamma\delta$ T17 cells (Figure 6F), consistent with a partial block in $\gamma\delta$ T17 maturation. *Id2* expression was markedly increased in all Id3-KO $\gamma\delta$ T cell subsets (Figure 6I), consistent with a previously reported compensatory role (Zhang, Lin, Dai, & Zhuang, 2014). *Maf* and *Zbtb16* transcripts were markedly reduced in Id3-KO cells at the $\gamma\delta$ Te stage but recovered to near WT levels in mature $\gamma\delta$ T17 cells (Figure 6I). We noted a similar impact on expression of *Rora* in Id3-KO mice. *Rora* is not required for $\gamma\delta$ T17 cell function (Barros-Martins et al., 2016), but does play roles in regulation the development of Th17 (Hall et al., 2022) and ILC3 cells (Lo et al., 2016). We also examined expression of *Rorc* (encodes ROR γ t) (Figure 6 – figure supplement 1). Given the decrease in other genes involved in $\gamma\delta$ T17 cell maturation, and the dependence of *Rorc* expression on MAF

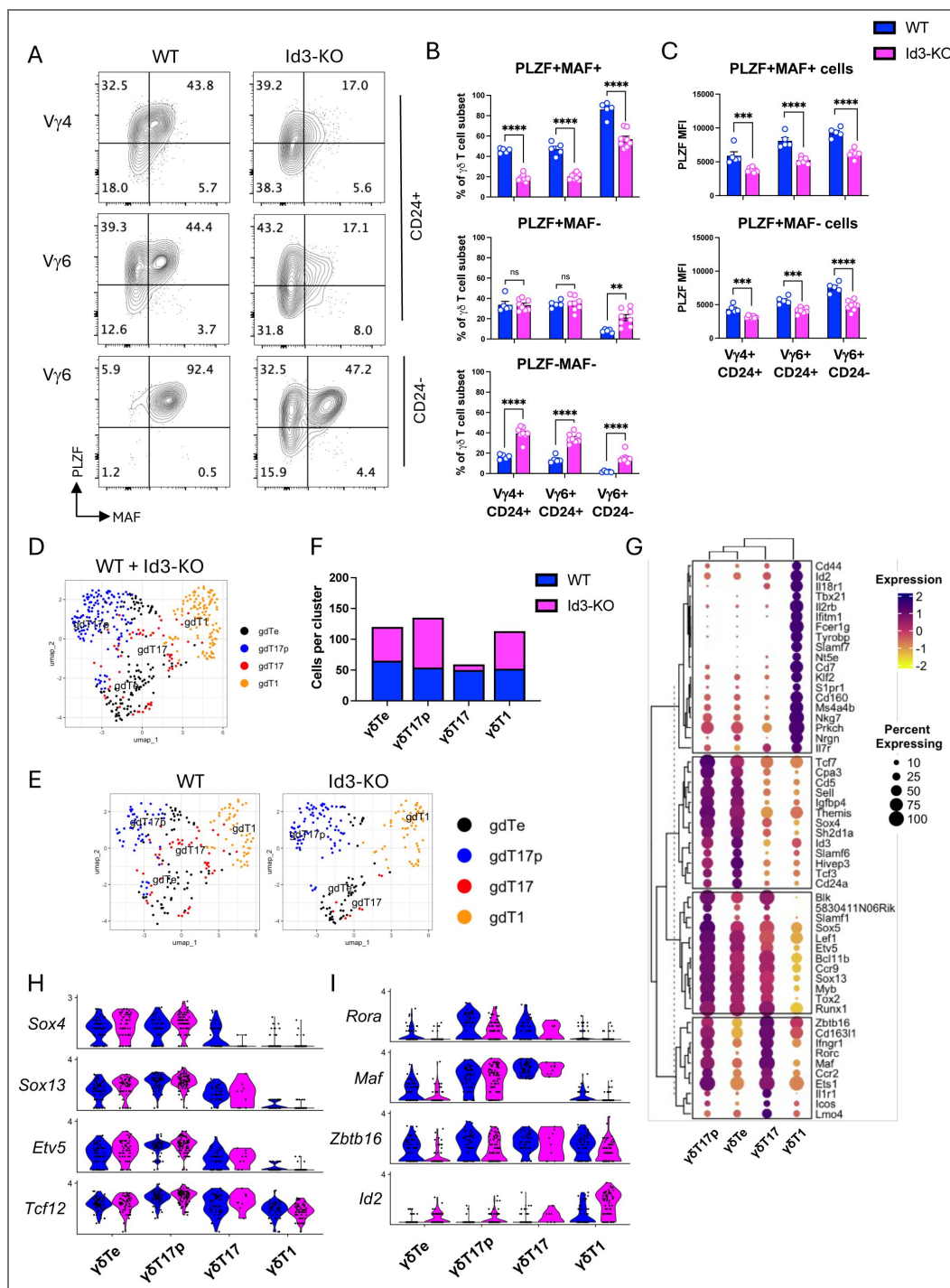


Figure 6. Intact $\gamma\delta$ T commitment gene program and impaired $\gamma\delta$ T17 maturation program in Id3-KO mice.

A. Flow cytometry plots showing the percentages of cells expressing PLZF and/or MAF in immature (CD24⁺) Vγ4 and Vγ6 cells, and in mature (CD24⁻) Vγ6 cells (note that mature Vγ4 cells are not present in the E18 fetal thymus). B. Quantification of the percentages of cells expressing PLZF and MAF (top), PLZF only (middle) or neither (bottom) within the immature Vγ4 and Vγ6 subsets, and the mature Vγ6 subset. C. Mean fluorescent intensities of PLZF in the PLZF⁺MAF⁺ and PLZF⁺MAF⁻ populations within the immature and mature Vγ subsets. D, E. sRNA-seq UMAP plots of $\gamma\delta$ T cells as merged (D) or split (E) into WT versus Id3-KO populations. F. Number of WT and Id3-KO cells per cluster. G. Clustered dot plot of curated gene sets used to assign $\gamma\delta$ T17p, $\gamma\delta$ Te, $\gamma\delta$ T17, and $\gamma\delta$ T1 identities. H. Expression of $\gamma\delta$ T cell commitment genes in WT versus Id3-KO cells by cluster. I. Expression of $\gamma\delta$ T17 maturation genes in WT versus Id3-KO cells by cluster. Blue = WT, pink = Id3-KO. ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

in $\gamma\delta$ T17 cells (Zuberbuehler et al., 2019 [↗](#)), we were surprised to find that *Rorc* was higher in Id3-KO $\gamma\delta$ T17p cells than in WT $\gamma\delta$ T17p cells (Figure 6 – figure supplement 1D) and was also elevated in the CD8-expressing DN4 subset (Figure 6 – figure supplement 1E). Although ROR γ t is a critical regulator of $\gamma\delta$ T17 development and function, it is also upregulated after β -selection and maintains DP cell survival (Xi, Schwartz, Engel, Murre, & Kersh, 2006 [↗](#)), suggesting that the increase we observed in the $\gamma\delta$ T17p cells could in part be due to $\alpha\beta$ -T cell lineage diversion.

ChIP-Seq analysis reveals shared HEB, E2A, and Egr2 binding sites in the *Id3* locus

Although our data showed that HEB is necessary for *Id3* expression during $\gamma\delta$ T17 development, it remained unclear whether HEB directly regulates *Id3*. The *Id3* gene locus is composed of three exons adjacent to the long non-coding RNA gene *Gm42329*, which lies in the opposite orientation (Figure 7A [↗](#)). To assess direct binding of HEB and/or E2A to the *Id3* locus, we analyzed ChIP-seq datasets from *Rag2*^{-/-} DN3 thymocytes (Fahl et al., 2021 [↗](#)). This analysis identified three major binding regions for both HEB and E2A upstream of the first exon of *Id3*, each containing multiple peaks. We also examined ChIP-seq datasets from DN3 cells stimulated with anti-CD3 or anti-TCR β to mimic TCR signaling (Pekowska et al., 2011 [↗](#); Seiler et al., 2012 [↗](#)). RNA polymerase II bound the *Id3* promoter in *Rag2*^{-/-} mice stimulated with anti-CD3 ϵ , revealing active *Id3* transcription in cells that had experienced CD3-mediated signaling. Additionally, we observed binding of Egr2 to two sites that overlapped with the HEB/E2A regions in total thymocytes from mice that had been injected with anti-TCR β (Seiler et al., 2012 [↗](#)). Notably, H3K27me3 repressive marks were detected across the *Gm42329* locus but were absent from *Id3*, indicating that *Id3* is epigenetically poised for activation before pre-TCR and/or $\gamma\delta$ TCR signaling occurs.

Given that the data from the Egr2 and HEB/E2A ChIP-seq analyses were generated using different thymocyte subsets, we analyzed additional ChIP-seq data for HEB and E2A binding in *Rag2*^{-/-} DN3 cells which had been transduced with retroviral constructs encoding the KN6 $\gamma\delta$ TCR and cultured with stroma expressing the weak KN6 ligand T10 for four days (Fahl et al., 2021 [↗](#)). These *Rag2*^{-/-} DN3 + $\gamma\delta$ TCR cells allowed us to examine HEB/E2A binding to sites in the *Id3* locus in a cellular context more closely aligned to the Egr2 binding assay (Figure 7 - figure supplement 1A). This analysis revealed that the binding of HEB/E2A on those sites persisted after weak $\gamma\delta$ TCR signaling, strengthening the likelihood that concurrent binding of HEB/E2A and Egr2 occurs during this developmental transition. We noted that HEB/E2A binding was slightly dampened in *Rag2*^{-/-} DN3 + $\gamma\delta$ TCR cells relative to *Rag2*^{-/-} DN3 cells, consistent with the induction of *Id3* and subsequent Id3-mediated disruption of E protein binding. We designated the regions of overlapping HEB, E2A, and Egr2 binding as *HE1* and *HE2*. Additionally, sequence-level analysis of *HE1* and *HE2* allowed the identification of predicted E protein binding sites (E box) in close proximity to Egr binding sites (Egr) (Figure 7-figure supplement 1B), suggesting that HEB/E2A and Egr2 may participate in an enhanceosome complex. Together, these findings support the hypothesis that HEB and E2A work in concert with Egr2 to modulate *Id3* transcription in thymocytes undergoing TCR-mediated selection.

To examine how the chromatin landscape of the *Id3* locus might change across this transition, we generated ATAC-seq data from DN3 and DN4 cells, as well as *Rag2*^{-/-} DN3 cells, to provide a genuine pre-selection context (Figure 7 – figure supplement 1A). Alignment of ATAC-seq and ChIP-seq peaks in the *Id3* locus revealed accessibility of *HE1* and *HE2* in *Rag2*^{-/-}, WT DN3, and WT DN4 cells, strengthening their relevance. Given the known ability of E2A and HEB to induce chromatin remodeling (Emmanuel et al., 2018 [↗](#); Lin et al., 2010 [↗](#)), we also examined accessibility in DN3 and DN4 cells from HEB cKO mice (Figure 7 – figure supplement 1A). Alignment of ATAC-seq and ChIP-seq peaks in the *Id3* locus revealed accessibility of *HE1* and *HE2* in *Rag2*^{-/-}, WT DN3, and WT DN4 cells. However, accessibility of *HE1* and *HE2* was dampened in HEB cKO cells, especially at the DN3 stage, suggesting that HEB may be involved in remodeling the *Id3* locus, resulting in a poised state that enables TCR-dependent transcription factors to induce *Id3* proportionally to TCR signal strength.

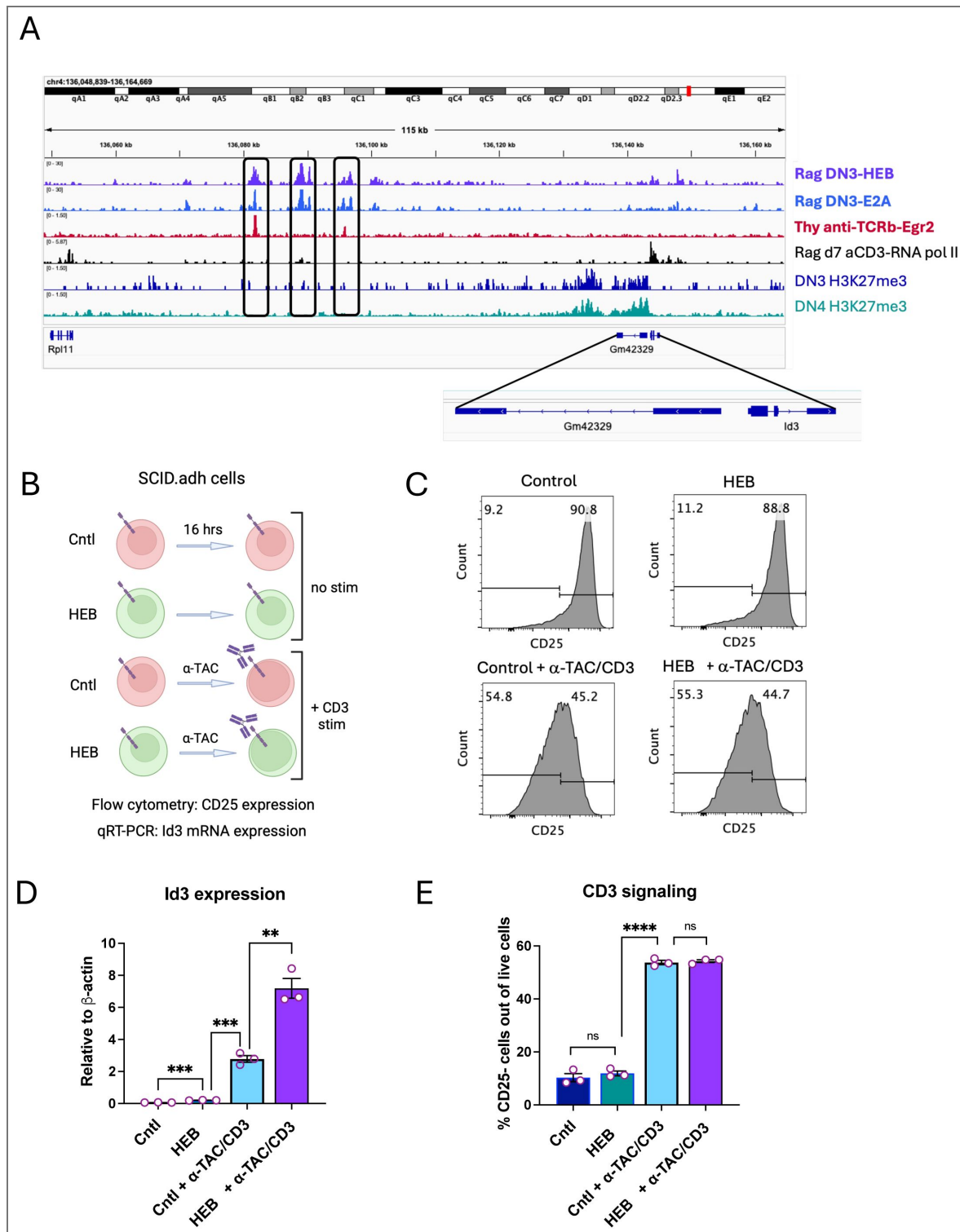


Figure 7. Synergistic upregulation of *Id3* by HEB and CD3 signaling.

A. ChIP-seq data analysis of the binding of HEB, E2A, RNA polymerase, and Egr2, and the extent of H3K27me3 chromatin modification, in DN3 and/or DN4 cells at the *Id3* gene locus, obtained from publicly available datasets (see Materials and Methods for accession numbers). The cell type and antibody used in each experiment are indicated to the right of the tracks. Peaks bound by HEB, E2A, and/or Egr2 are indicated in boxes. Inset shows the *Id3* exons and the adjacent *Gm42329* long non-coding RNA. B. Diagram of experimental design. SCID.adh cells transduced with HEBAlt or control retroviral vectors were cultured for 16 hours in the presence or absence of the anti-TAC antibody, which induces signaling through the CD3 complex. C, E. Flow cytometry plots (C) and quantification (E) of CD25 upregulation with and without stimulation and/or HEB expression. D. *Id3* mRNA expression relative to β -actin as determined by quantitative RT-PCR. Rag = *Rag2*^{-/-} mouse thymocytes, which are arrested at the DN3 stage of development. ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

TCR Signaling and HEB Converge to Potentiate *Id3* Transcription

To test whether HEB can cooperate with TCR/CD3 signaling to upregulate *Id3*, we used a gain-of-function approach. We took advantage of the SCID.adh cell line, which ectopically expresses a chimeric hIL-2R α :CD3 ϵ receptor that mimics pre-TCR signaling when stimulated with anti-TAC (hIL-2R α) antibody (Carleton et al., 1999). Stimulation results in downregulation of CD25, *Rag1*, *Rag2*, and *Ptcra*, while inducing *Trac* germline transcripts, recapitulating pre-TCR activity. We transduced SCID.adh cells with either control or HEB-expressing retroviruses to generate control and HEB-overexpressing cells (Figure 7B). To avoid the growth arrest and cell death associated with full-length HEB (HEBCan) overexpression (Engel & Murre, 2004; Wang et al., 2010), we used a construct encoding HEBAlt, a truncated form that activates E protein target genes without impairing cell viability (Wang et al., 2010; Wang et al., 2006; Yoganathan et al., 2022).

Upon overnight stimulation with anti-TAC, both control and HEB-expressing cells downregulated CD25 to similar degrees, as assessed by flow cytometry, confirming effective CD3-mediated signaling (Figure 7C, E). Quantitative RT-PCR analysis showed that *Id3* expression was modestly elevated in unstimulated HEB-expressing cells compared to controls, and stimulation of control cells upregulated *Id3* levels to a greater degree (Figure 7D). Notably, *Id3* expression in stimulated cells expressing HEB was much higher than either HEB or stimulation alone, and exceeded the sum of these two conditions, indicating a synergistic interaction. These results demonstrate that HEB can amplify *Id3* induction in response to TCR signaling, potentially through cooperative interaction with TCR-induced factors such as Egr2.

Discussion

HEB and *Id3* are both critical for T cell development, but their distinct functions in fetal $\gamma\delta$ T cell development have not been resolved. Here, we show that HEB controls expression of genes involved in $\gamma\delta$ TCR signaling and induces a transcriptional program that primes $\gamma\delta$ T cell precursors for $\gamma\delta$ T17 differentiation. Furthermore, HEB collaborates with TCR-dependent factors to upregulate *Id3*, which enables $\gamma\delta$ T17 cell maturation and inhibits the $\alpha\beta$ T cell fate. Together, these findings define a sequence of regulatory states governed by the E/*Id* axis that orchestrate $\gamma\delta$ T cell lineage commitment and the differentiation of functional $\gamma\delta$ T17 cells.

TCR signal strength is tightly linked to specific combinations of V γ and V δ chains expressed on fetal $\gamma\delta$ T cells, which serve as critical drivers of $\gamma\delta$ T cell fate and functional programming (Chen et al., 2021; Fahl, Kappes, et al., 2018; Munoz-Ruiz et al., 2016; Scaramuzzino et al., 2022). We found that HEB is required for expression of TRGV4 and TRDV5. These genes encode the V γ 4⁺V δ 5⁺ TCR, which supports $\gamma\delta$ T17 cell differentiation (Kashani et al., 2015). While TRGV4 is known to be regulated by E proteins (Bain, Romanow, Albers, Havran, & Murre, 1999; In et al., 2017; Nozaki et al., 2011), the requirement for HEB in TRDV5 expression is newly appreciated. Notably, the TRD locus contains E2A-dependent insulators that limit TRDV4 expression to the fetal thymus (B. Hao & Krangel, 2011). Whether these are also HEB-dependent, and whether they impact TRDV expression among fetal $\gamma\delta$ T cell subsets, remains to be determined.

Our analysis identified known E protein target genes, including components of the TCR signaling pathway (*Cd3d*, *Cd3g*, *Lat*, *Zap70*) (Braunstein & Anderson, 2011; M. Miyazaki et al., 2017) and chemokine receptors (*Cxcr5*, *Cxcr4*, *Ccr9*) (Brauer et al.), validating the experimental approach. Moreover, we found that a suite of $\gamma\delta$ T17-specific genes were also HEB-dependent, including *Blk* and *Syk*, which are $\gamma\delta$ T17 cell-specific mediators of TCR signaling. We also noted that several inhibitors of TCR signaling were decreased, including *Pdcd1* (encodes PD-1), *Nfya* (encodes Ikappa α), and *Sh2d2a* (encodes TSAd). All of these inhibitors are regulated post-translationally, suggesting a role for HEB in inducing T-lineage specific pathways that enable both positive and negative regulation of TCR signal transduction.

Our studies show that *Id3* limits $\alpha\beta$ -lineage potential even in the presence of HEB, suggesting that HEB restricts the $\alpha\beta$ program primarily by upregulating *Id3*. In contrast, the early $\gamma\delta$ T cell program is upregulated in *Id3*-KO mice, indicating that *Id3* is not required for this function. A third E/*Id* dynamic operates during $\gamma\delta$ T17 maturation, which requires *Id3* independently of HEB. Sox13

has been shown to initiate a cascade of regulatory events that induce *Blk* and *Maf* during $\gamma\delta$ T17 development (In et al., 2017 [DOI](#); Sagar et al., 2020 [DOI](#)). However, our work indicates that *Sox13* is not sufficient to upregulate $\gamma\delta$ T17 maturation factors like *Zbtb16* and *Maf* in the absence of *Id3*, implying an E-protein dependent brake on $\gamma\delta$ T17 maturation. This highlights a decoupling between lineage specification and effector maturation, supporting the two-step model of $\gamma\delta$ T cell development (Buus et al., 2016 [DOI](#)).

HEB and E2A can partially compensate for each other in the context of lymphocyte development, complicating interpretations of single gene knockouts (R. J. Barndt, Dai, & Zhuang, 2000 [DOI](#); Zhuang, Barndt, Pan, Kelley, & Dai, 1998 [DOI](#)). This has been especially well characterized in $\alpha\beta$ -T cell development, in which a lack of E2A leads to partial blocks at the ETP and DP to SP transitions (Ikawa, Kawamoto, Goldrath, & Murre, 2006 [DOI](#); Jones & Zhuang, 2007 [DOI](#)), whereas HEB deficiency impairs the ISP to DP transition (R. Barndt et al., 1999 [DOI](#)). Mice conditionally lacking both HEB and E2A have more severe phenotypes (Jones & Zhuang, 2011 [DOI](#)), especially at the ETP to DN2 transition when T-cell lineage identity is being established (M. Miyazaki et al., 2017 [DOI](#)), and at the DN2 to DN3 transition, when T-lineage commitment occurs (Wojciechowski et al., 2007 [DOI](#)). Indeed, early loss of HEB alone renders the T cell precursors more open to divergence to non-T lineages, even at the DN3 stage (Braunstein & Anderson, 2011 [DOI](#)). Our laboratory and others identified “DN1-like” cells in the thymus of E protein deficient and *Id1*-overexpressing mice, which were later identified as ILC2s (Berrett et al., 2019 [DOI](#); K. Miyazaki et al., 2025 [DOI](#); M. Miyazaki et al., 2017 [DOI](#); Qian et al., 2019 [DOI](#)). The TCR γ rearrangements observed in thymic ILC2s and peripheral ILC2s in normal mice (Pankow & Sun, 2022 [DOI](#); Qian et al., 2019 [DOI](#); Shin et al., 2020 [DOI](#)), support a critical role for E proteins in lineage divergence, rather than outgrowth of alternative lineages, even prior to the $\alpha\beta/\gamma\delta$ T cell branchpoint.

Our findings are consistent with a selective requirement for HEB factors in $\gamma\delta$ T17 cell differentiation. Intriguingly, we observed that $\gamma\delta$ T1 lineage cells express *Tcf3* (E2A) and *Id3* but not *Tcf12* (HEB). Furthermore, *Tcf3* levels remained constant under conditions of HEB deficiency in all $\gamma\delta$ T cell subsets. These results suggest that while E2A is insufficient for $\gamma\delta$ T17 development, it may be required and sufficient for $\gamma\delta$ T1 development. Overall, our analysis suggests a strong division of labor between HEB and E2A in promoting the $\gamma\delta$ T17 and $\gamma\delta$ T1 fates, respectively.

It is possible that *Id3* is needed after $\gamma\delta$ T cell specification to inhibit E2A-driven inhibitors of $\gamma\delta$ T17 maturation. Possible candidates include positive regulators of $\gamma\delta$ T1 differentiation, such as T-bet (*Tbx21*) or Eomes. These transcription factors participate in negative cross-regulatory loops with $\gamma\delta$ T17 regulators such as Runx1, ROR γ t, and AP-1 factors, which help to stabilize $\gamma\delta$ T cell subset lineage identity (Parker et al., 2025 [DOI](#)). Additionally, transiently expressed HEB-dependent transcription factors such as *Etv5* and *Sox5* may act as a checkpoint for $\gamma\delta$ T17 maturation, to be released upon *Id3* upregulation.

The TCR signal strength model posits that strong signals induce high levels of Egr factors, which induce high levels of *Id3* (Haks et al., 2005 [DOI](#); Lauritsen et al., 2009 [DOI](#)). Our data add an important new dimension to this paradigm, proposing that the lower Egr2 levels induced by weaker $\gamma\delta$ TCR signaling require cooperation with HEB/E2A to upregulate *Id3* expression (Figure 7 - figure supplement 2A). This model is supported by our ChIP-seq data, which confirms HEB and E2A binding at the *Id3* locus, at regions also bound by Egr2, and gain-of-function studies, which show that HEB synergizes with TCR/CD3 signals to amplify *Id3* expression. However, it is important to note that additional studies will be needed to confirm whether HEB and E2A collaborate with Egr2 in a temporally coordinated fashion during $\gamma\delta$ T cell specification. Both HEB and E2A are pioneering factors and can increase locus accessibility of genes involved in lymphoid development, such as *Foxo1* (Welinder et al., 2011 [DOI](#)). Interestingly, we found that HEB/E2A peaks in the *Id3* locus are diminished in cells from HEB cKO mice relative to WT, particularly at the DN3 stage. Therefore, HEB may play an important role in epigenetically priming the *Id3* locus prior to TCR signaling.

Our data is consistent with a partial compensation for *Id3* by *Id2*, less effectively during $\gamma\delta$ T cell commitment, and more fully during late $\gamma\delta$ T17 maturation, when *Id2* is normally expressed. This is consistent with the observation that *Id3* supports transient developmental transitions, while *Id2*

stabilizes innate-like transcriptional states (Anderson, 2022 [↗](#)). Since *Id2* is a direct target of E2A (Schwartz, Engel, Fallahi-Sichani, Petrie, & Murre, 2006 [↗](#)), the upregulation of *Id2* in *Id3*-deficient precursors likely reflects increased E protein activity. Thus, HEB, *Id3*, and *Id2* participate in negative regulatory loops that allow transient HEB activity and *Id3* expression, followed by stabilization of the innate $\gamma\delta$ T17 gene network by *Id2* (Figure 7 – figure supplement 2B). Understanding how *Id3* and *Id2* differentially regulate the timing and magnitude of E protein activity, and whether they have different impacts on E2A/E2A homodimers versus HEB/E2A heterodimers remains to be addressed.

There are limitations to our study. Our analyses focused on E18 $\gamma\delta$ T cells, which reflect $\gamma\delta$ T17-biased fetal development and may not capture functions of HEB or *Id3* at other stages. We previously showed that HEB cKO mice have defects in the production of functional $\gamma\delta$ T17 cells in fetal and neonatal thymus (In et al., 2017 [↗](#)). We also found that $\gamma\delta$ T cells from HEB cKO mice exhibited a diminished capacity for IL-17 production in adult lungs and spleen $\gamma\delta$ T cells. While the adult thymus does not support the development of fully functional innate $\gamma\delta$ T cells (Chen et al., 2021 [↗](#); J. D. Haas et al., 2012 [↗](#); Kashani et al., 2015 [↗](#)), it does contain $\gamma\delta$ TCR⁺ cells that have activated *Sox-Maf-Rorc* gene networks (Yang et al., 2023 [↗](#)). In addition, we have not yet addressed the roles of HEB and *Id3* in TCR-independent development of a subset of early fetal-derived V γ 4⁺ $\gamma\delta$ T cells (Spidale et al., 2018 [↗](#)). Importantly, the specific contributions of HEBAlt and HEBCan isoforms remain unresolved, as both were deleted in our conditional HEB model. Further studies of $\gamma\delta$ T cell development in the neonatal and adult thymus of HEB cKO as well as HEBAlt KO and HEBCan KO mice are underway.

In summary, our studies have identified multiple interlinked transcriptional circuits that require E proteins and *Id* factors during $\gamma\delta$ T cell development. HEB induces *Id3*, which then inhibits E protein activity. In the absence of *Id3*, HEB and E2A activity persist, inducing compensatory *Id2* expression. Since *Id3* expression is self-limiting through E protein suppression, the HEB-*Id3* interactions result in a negative feedback loop. Thus, HEB plays dual roles in establishing $\gamma\delta$ lineage identity and initiating $\gamma\delta$ T17 differentiation via *Id3*. Future work should clarify the direct transcriptional targets and co-factors of HEB, as well as how dynamic levels of HEB, *Id3*, and *Id2* coordinate $\gamma\delta$ T cell fate decisions.

Materials and methods

Experimental Design and Statistical Analysis

The overall goal of this study was to understand how HEB transcription factors regulate the transcriptional networks required for the development of IL-17 producing $\gamma\delta$ T cells. The subjects of these studies were genetically modified mice. The experimental variable was the genotype of the mice, chosen based on availability of animals and littermate controls. For the scRNA-seq experiments, fetal thymocytes of like genotype from 3 or more mice were pooled prior to enrichment. Each flow cytometry analysis was performed using 3 or more mice of like genotype per experiment, repeated 2-3 times. Mice of both sexes were used. No sex differences were apparent; therefore, data were pooled. Each data point depicted in graphs represents an individual mouse. The investigators were not blinded to allocation during experiments and outcome assessment, except when fetal thymocytes were individually analyzed by flow cytometry prior to genotyping. Significance of pairwise data was computed by unpaired two-tailed Student's t-test, with $p < 0.05$ considered statistically significance. The error bars represent standard error of the mean (SEM) values.

Mice

HEB^{fl/fl} mice have loxP sites flanking exons that encode the bHLH DNA binding and dimerization domain, which is shared by all HEB isoforms (Wang et al., 2006 [↗](#); Wojciechowski et al., 2007 [↗](#)). These mice were bred to Vav-iCre transgenic mice (JAX stock #008610) (de Boer et al., 2003 [↗](#)), which deletes regions flanked by loxP sites in all hematopoietic cells (Siegemund, Shepherd, Xiao, & Sauer, 2015 [↗](#)), to generate HEB^{fl/fl;Vav-iCre} (HEB cKO) mice, as previously described (In et al.,

2017 [\[1\]](#); Welinder et al., 2011 [\[2\]](#)). All experimental mice were obtained from timed matings of HEB^{fl/fl} x HEB^{fl/fl};Vav-iCre mice, giving rise to WT and HEB cKO littermates for experimental use. *Id3* deficient mice (*Id3*-KO) lacked *Id3* in all cells due to a knock-in/knock-out allele in which the *Id3* coding sequence was replaced with red fluorescent proteins (RFP) (JAX strain # 010983, B6;129S-*Id3*^{tm1Pzg/J}). RFP was not detectable in our analyses due to quenching during intracellular staining. *Id3*-KO mice were acquired on the E29/B6 background and bred onto the C57Bl/6 background for 8 generations. *Id3*^{+/-} mice were timed mated together to produce WT and *Id3*-KO littermates. Mice were bred and maintained in the Comparative Research Facility of the Sunnybrook Research Institute (Toronto, Ontario, Canada) under specific pathogen-free conditions. All animal procedures were approved by the Sunnybrook Research Institute Animal Care Committee.

Timed Matings and Embryo Harvest

Timed matings were performed by setting up mating pairs (day 0) and separating them ~16 hrs later. Embryos were harvested after 18 days (E18). Fetal thymuses were dissected, and single cell suspensions were generated by mechanical disruption and filtered through 40-micron mesh. Cells were resuspended in 1X HBSS/BSA for sorting or flow cytometry. Embryos were processed and analyzed separately, and tail tissue was collected for genotyping of fetal mice. Genotyping was performed as previously described (Wojciechowski et al., 2007 [\[3\]](#)).

Flow cytometry

Antibodies were purchased from eBiosciences (San Diego, California, USA), Biolegend (San Diego, California, USA) and BD Biosciences (Mississauga, Ontario, Canada). For flow cytometry, cells were washed and incubated with Fc blocking antibody (BD Biosciences), followed by extracellular staining for surface CD4 (clone GK1.5), CD8α (clone 53–6.7), CD3 (clone 145-2C11), TCRγδ (clone GL3), Vγ4 (Vγ2; clone UC3-10A6), Vγ5 (Vγ3; clone 536), Vγ1 (Vγ1.1; clone 2.11), CD27 (clone LG3-1A10), CD24 (clone M1/69), CD73 (clone TY11.8), and CD25 (clone PC615). To assess functional capacities, cells were stimulated by incubation for four h with PMA (50 ng/ml) and Ionomycin (500 ng/ml) in the presence of Brefeldin A (5mg/ml; eBioscience), washed with 1X HBSS/BSA, and incubated with Fc block before staining for surface epitopes. Cells were then fixed and permeabilized (Fix and Perm Cell Permeabilization Kit; eBioscience) and stained with antibodies against IL-17A (clone eBio17B7). For intracellular transcription factor staining, the cells were fixed, permeabilized (FoxP3 Staining Kit, eBioscience) and stained for MAF (clone sym0F1), and PLZF (clone R17-809). All flow cytometric analyses were performed using Becton-Dickenson (BD) LSRII, Fortessa, or SymphonyA5-SE cytometers. FACSDiva and FlowJo software were used for analysis. Sorting was performed on BD ARIA and BD Fusion sorters.

Cell culture

SCID.adh cells, which have been engineered to express a surface human IL-2Ra (TAC):CD3ε chimeric signaling molecule on the surface, were cultured as previously described (Anderson, Weiss, Hernandez-Hoyos, Dionne, & Rothenberg, 2002 [\[4\]](#)). Cells were transduced with MIGR1 encoding GFP only, or MIGR1 encoding GFP and HEBA1. GFP⁺ cells were sorted, expanded, and cultured overnight on plates coated with anti-TAC antibody at 5μg/mL in 500 μl of PBS, or PBS only as control. Cells were analyzed by flow cytometry for expression of CD25, and for mRNA expression of *Id3* by qRT-PCR.

RNA extraction and qRT-PCR

Total RNA was extracted from cells using TRIzol® Reagent (Invitrogen) and reverse-transcribed into complementary DNA using Superscript III (Invitrogen). Reactions for qRT-PCR were prepared using PowerUp SYBR™ Green Master Mix (ThermoFisher) or Luna® Universal qPCR Master Mix (NEB) and 0.5 μM of primers. Primers (5' to 3') were as follows: b-actin forward: ATGGTGGGAATGG GTCAGAA, b-actin reverse: TCTCCATGTCGTCCAGTTG, *Id3* forward: CTGTCGGAACGTAGCCTGG, *Id3* reverse: GTGGTTCATGTCGTCGAAGAG. The qRT-PCR reaction was run and analyzed using an

Applied Biosystems 7500 Fast Real-Time PCR System (ThermoFisher) and a QuantStudio™ 5 Real-Time PCR System (ThermoFisher). Values from the qRT-PCR reactions were normalized to β -actin values, and relative expression values were calculated by the delta Ct method.

Single-cell RNA sequencing (scRNA-seq) of WT and HEB cKO $\gamma\delta$ T cells

E18 fetal thymuses from HEB cKO mice were dissected from embryos and individually pressed through a 40-micron mesh with a syringe plunger to create single cell suspensions. Cells were incubated with Fc block for 30 min on ice. Small aliquots from each sample were stained with CD4, CD8, and CD3 to assess genotypes, with a lack of DP indicating HEB deletion, and genotypes were verified by PCR. Five WT and five HEB cKO littermates were pooled by genotype, stained, and sorted for $\text{TCR}\gamma\delta^+\text{CD}3^+$ cells. Samples were not hash-tagged as this technology was not available at the time of the experiments. Fifty thousand cells per sample were loaded onto the 10X Chromium controller, and barcoded libraries were generated using the Chromium Next GEM Single Cell 5' Kit v2 (Dual Index) (10X Genomics) at the Princess Margaret Genomics Facility (Toronto, ON, Canada). Next generation Illumina sequencing was performed to a depth of $\sim 30,000$ reads per cell. The estimated number of cells sequenced was ~ 3300 for WT and ~ 5500 for HEB cKO samples.

scRNA-seq of WT and Id3-KO DN fetal thymocytes

E18 fetal thymuses from Id3-KO mice were dissected from embryos, pressed through mesh as above, and incubated with Fc block. Embryos were genotyped by PCR, and thymocytes were pooled according to genotype (3 WT and 3 Id3-KO mice). DN cells were enriched by magnetic sorting using anti-CD4 and anti-CD8 microbeads according to manufacturer's instructions (Miltenyi Biotech). Flow-through ($\text{CD}4^-\text{CD}8^-$) cells were processed in-house using the Chromium Next GEM Single Cell 3' Kit v2 (Dual Index) (10X Genomics). Next generation sequencing was performed using the Illumina platform to a depth of $\sim 100,000$ reads per cell. The estimated numbers of cells sequenced for each sample was ~ 3700 .

scRNA-seq data analysis of WT and HEB cKO $\gamma\delta$ T cells

FASTQ raw data files were aligned to the mm10 genome using the STAR aligner (STAR v2.5.2b). Cell Ranger (v1.1.7; 10X Genomics) (G. X. Zheng et al., 2017 [DOI](#)) was used to generate matrix files which were analyzed using programs in R Seurat version 4.4 (Y. Hao et al., 2021 [DOI](#)), as detailed in the R-Markdown files. WT and HEB cKO datasets were processed for quality control by excluding cells with more than 7% mitochondrial genes, less than 1000 unique genes, and/or less than 4000 transcripts, resulting in 1272 WT cells and 1951 HEB cKO cells for further analysis. Filtered WT and HEB cKO datasets were merged, and SCTransform was applied to the merged dataset. Cell cycle regression was performed to mitigate the influence of cell cycle heterogeneity on clustering, and cells expressing high levels of *Lyz2* were excluded to remove most myeloid cells. PCA analysis was performed (RunPCA) and used to compute a nearest neighbor graph (FindNeighbors) and identify clusters (FindClusters). UMAP plots were generated using RunUMAP and displayed using DimPlot_scCustom. FindMarkers was used to identify the top ten most differentially expressed genes between clusters which were visualized by heatmap (DoHeatmap).

Generation of gene lists and module scores

To construct an unbiased and comprehensive list of genes diagnostic for different stages and lineages of $\gamma\delta$ T cell development, we collected lists of differentially expressed genes from eight publications characterizing $\gamma\delta$ T cell subsets in the fetal thymus, adult thymus, and adult peripheral tissues (du Halgouet et al., 2024 [DOI](#); Inacio et al., 2025 [DOI](#); Liu et al., 2020 [DOI](#); Mistri et al., 2024 [DOI](#); Narayan et al., 2012 [DOI](#); Sagar et al., 2020 [DOI](#); Spidale et al., 2018 [DOI](#); Yang et al., 2023 [DOI](#)). Genes were filtered to remove those involved in cell cycle and metabolism, as well as NK cell receptor genes. Lists were combined, duplicates were removed, and a final list of 87 genes was obtained (Figure 2 – figure supplement 1). This list was used for computation of the top ten genes per cluster that were most differentially expressed from all other clusters, visualized as a

Clustered DotPlot. Comparisons with the source literature and the Immunological Genome Project (Heng, Painter, & Immunological Genome Project, 2008 [\[1\]](#)) were used to categorize the clusters. $\gamma\delta$ Te1 and $\gamma\delta$ Te2 represent two types of early $\gamma\delta$ T cell subsets with random assignments into “1” and “2”.

As developmental stages are continuous, not discrete, some genes were present in more than one module; however, scores were based on all genes in each module. Genes in each module are listed below.

gdTe1: Ccr9, Sox13, Cpa3, Etv5, Sox5, Tox2, Ccr2, Igfbp4, Blk, Lmo4, Slamf1, Ifngr1, Rorc, Maf, Icos
gdTe2: Hivep3, Cd28, Themis, Slamf6, Sell, Igfbp4, Cd24a, Gzma
gdT17p: Blk, Il17re, Vdr, Il17a, Lmo4, Slamf1, Ifngr1, Rorc, Maf, Icos
gdT17: Il18r1, Il7r, Id2, Il23r, Il1r1, Cd44
gdT1p: Klf2, Nrgn, Cd2, Ms4a4b, Prkch, Nr4a1, Id3
gdT1: S1pr1, Eomes, Slamf7, Tyrobp, Ifitm1, Fcer1g, Tbx21, Gzmb
abT: Cd8a, Notch1, Cd8b1, Rag1, Rag2, Rmnd5a

Module location and intensity in WT versus HEB cKO cells were visualized using split UMAPs. Comparisons of the expression of single genes between WT and HEB cKO clusters were shown using split violin plots. To compare Clusters 1 and 4, we performed FindMarkers using the Wilcox test, with a minimum of three cells per feature and three cells per group. Differentially expressed genes were displayed using an Enhanced Volcano Plot, with significance set at $\log_2$ Fold change > 0.5 and $-\log_{10}P < 10^{-26}$.

scRNA-seq data analysis of WT and Id3-KO DN thymocytes

The following analyses were conducted on the Id3-KO dataset. FASTQ raw data files were aligned to the mm39 genome using the STAR aligner (STAR v2.5.2b), and Cell Ranger was used to generate matrix files, which were analyzed using programs in R-Seurat. Cells were computationally filtered as described above in the HEB cKO dataset, resulting in 2323 WT cells and 2893 Id3-KO cells for further analysis. Datasets were merged and subjected to SCTransform as above. PCA plots were generated and used to construct UMAP plots depicting clusters in merged datasets and distribution of WT versus Id3-KO cells within each cluster by UMAP (see R-markdown file). Feature plots and violin plots were used to identify clusters with $\gamma\delta$ T cell characteristics, which were subsetted using FindClusters. ClusteredDotPlot was used to visualize the top eight most differentially expressed genes within the curated gene set described above. Split violin plots were generated to show relative expression of early and late $\gamma\delta$ T cell genes in WT versus HEB cKO cells within each cluster.

ATAC-seq

Thymuses were dissected from adult Rag2^{-/-}, WT and HEB cKO mice, pressed through mesh as above, and incubated with Fc block. WT and HEB cKO DN cells were enriched by magnetic sorting using anti-CD4 and anti-CD8 microbeads according to manufacturer’s instructions (Miltenyi Biotech). Flow-through (CD4⁺CD8⁻) cells were stained and flow sorted into two populations: DN3 (CD4⁺CD8⁻CD44⁺CD25⁺) cells and DN4 (CD4⁺CD8⁻CD44⁺CD25⁻). DN3 cells were sorted from unfractionated Rag2^{-/-} thymocytes, which do not develop past the DN3 stage (Shinkai et al., 1992 [\[2\]](#)). Duplicates were generated for each subset, with each replicate derived from three mice. Replicates were sorted on different days. Both males and females were used. Sorted cells were cryopreserved in 50% FBS/40% growth media/10% DMSO in aliquots of 100,000 cells, which were subjected to ATAC-sequencing as follows. After membrane permeabilization, a transposase loaded with sequencing adaptors was added, which mediated insertion of adaptors at accessible genomic locations. Libraries were amplified and subjected to paired end next-generation sequencing using the Illumina platform. Paired end ATAC-seq libraries were generated and sequenced by Active Motif (Carlsbad, California).

ATAC-seq analysis

Bcl2fastq2 (v2.20) was used for processing of Illumina base-call data and demultiplexing, and bwa (v0.7.12) was used to align of reads to reference genome (mm10). Samtools (v0.1.19) was used to process BAM files. BEDtools (v2.25.0) was used to process of BED files, and wigToBigWig (v4) was used for generation of bigWIG files. bigWIG files were aligned using the Integrative Genomics Viewer (IGV) software (v2.16.2). Each sample yielded ~65 million mapped reads.

Gene ontology analysis

Genes differentially expressed between Cluster 1 and Cluster 4 with a significance of $\text{Log}_2\text{Fc} > 0.25$ and adjusted P value < 0.001 (Figure 4 – fig supplement 1) were submitted to ShinyGO 8.0 (Ge, Jung, & Yao, 2020) for gene ontology analysis. Pathway inclusion was set at a minimum of 5 genes with a false discovery rate (FDR) of 0.05. Genes were analyzed using the Kyoto Encyclopedia of Genes and Genomics (KEGG) pathway database (Kanehisa, Furumichi, Sato, Matsuura, & Ishiguro-Watanabe, 2025) (Figure 4 – figure supplement 1). Results were visualized as a bar graph showing Fold Enrichment (numbers) and $-\log_{10}(\text{FDR})$ values (colors).

Alignment of ChIP-seq data

ChIP-seq data for Rag DN3 and Rag DN3 + $\gamma\delta\text{TCR}$ cells binding to HEB and E2A were generated as follows: Rag2^{-/-} fetal liver hematopoietic cells were cultured on OP9-DL4 stroma to produce DN3 cells. Rag2^{-/-} DN3 cells were retrovirally transduced with the KN6 $\gamma\delta\text{TCR}$ or control (no TCR), and sorted GFP⁺ DN3 cells were cultured for four days on stroma expressing the weak KN6 ligand T10 to initiate $\gamma\delta\text{TCR}$ signaling. These cells were subjected to ChIP-seq using anti-E2A and anti-HEB antibodies as previously described (Fahl et al., 2021). All other files were obtained from the Cistrome database (R. Zheng et al., 2019) and aligned to the mouse genome (mm38) using the Integrated Genomics Viewer (IGV) (Robinson et al., 2011). Sources were as follows: Thy anti-TCRb-Egr2, GEO accession # GSM845900 (Seiler et al., 2012); Rag d7 aCD3-RNA pol II, GEO accession # GSM1340642 (Pekowska et al., 2011); DN3 H3K27me3, GEO accession # GSM1498423, and DN4 H3K27me3, GEO accession # GSM1498422 (Oravec et al., 2015).

Data availability

Matrix files and R-markdown code have been deposited to the Dryad repository and can be accessed private for review at:

HEB-WT scRNA-seq matrix files and R-markdown code

<https://doi.org/10.5061/dryad.08kpr5cq>

HEB cKO scRNA-seq matrix files

<http://doi.org/10.5061/dryad.qv9s4mwqh>

Id3-WT scRNA-seq matrix files and R-markdown code

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Id3-KO scRNA-seq matrix files

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ATAC-seq bigwig files are available at Dryad:

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Additional files

Figure 1 - figure supplement 1. Defects in $\alpha\beta$ T cell development in E18 fetal thymus of HEB cKO mice. Thymocytes were dissected from E18 WT and HEB cKO littermates and subjected to flow cytometry. At E18 very few cells had become CD4 or CD8 single positive cells with most cells at the DP stage in WT mice. The DN to ISP and ISP to DP transitions were severely compromised in the HEB cKO fetal thymus, in agreement with previously reports of HEB-deficient adult mice.

Figure 2 - figure supplement 1. References for curated gene list used to identify differentially expressed genes and assign cluster identities in Figure 2 and Figure 6. See spreadsheet.

Figure 2 - figure supplement 2. TRGV and TRDV expression profiling reveals depletion of TRGV4 and TRDV5 transcripts and overexpression of TRDV4 in the HEB cKO fetal $\gamma\delta$ T cells. **A.** Violin plots of expression of canonical genes that mark $\gamma\delta$ T cell subsets. **B.** Violin plots showing expression of TRGV and TRDV genes in WT versus HEB cKO by cluster. WT = blue, HEB cKO = orange. **C.** Blended split feature plots showing cells expressing TRGV chains (blue) or TRDV chains (red), and cells co-expressing TRGV and TRDV chains (pink). Co-expression in WT cells are shown on the top panel of each comparison and HEB cKO cells are shown on the bottom.

Figure 4 - figure supplement 1. Differentially expressed genes between WT and HEB cKO $\gamma\delta$ T17 precursor populations from Figure 4A. See spreadsheet.

Figure 4 - figure supplement 2. Enriched KEGG pathways and gene members between WT and HEB cKO $\gamma\delta$ T17 precursor populations, as shown in Fig. 4B. See spreadsheet.

Figure 4 - figure supplement 3. Patterns of E protein and Id protein gene expression during $\gamma\delta$ T cell development in WT and HEB cKO mice. **A.** Relative expression of *Tcf12* (HEB), *Tcf3* (E2A), *Id3*, and *Id2* in WT versus HEB cKO cells by cluster. WT = blue, HEB cKO = orange. **C-F.** Co-expression of E protein and Id protein transcripts assessed by blended split feature plots, for (B) *Tcf12* and *Tcf3*, (C) *Tcf12* and *Id3*, (D) *Tcf3* and *Id3*, (E) *Tcf12* and *Id2*, and (F) *Tcf3* and *Id2*. Blue = E protein gene expression, red = Id gene expression, pink = co-expression.

Figure 5 - figure supplement 1. CD73 is upregulated during development of $V\gamma 5$ and $V\gamma 1$ $\gamma\delta$ T cells in WT but not Id3-KO fetal thymus. Flow cytometry plots of CD24 and CD73 expression within the $V\gamma$ subsets from WT and Id3-KO E18 fetal thymus.

Figure 6 - figure supplement 1. Figure 6 – figure supplement 1. Identification of $\gamma\delta$ T cell subsets from E18 WT and Id3-KO DN cells using scRNA-seq. WT and Id3-KO E18 thymocytes were pooled and subjected to magnetic sorting to obtain CD4⁺CD8⁺ (DN) cells for scRNA-seq. **A.** UMAP of merged dataset depicting 11 clusters (0-10). **B.** Expression of lineage-defining genes to assign identities to clusters in merged dataset: *Cd3e* for T-lineage, *Spi1* (encodes PU.1) for myeloid lineage, *Sox13* for $\gamma\delta$ T17 lineage, *Maf* for myeloid and $\gamma\delta$ T17 lineages, and *Il2rb* and *Xcl1* for $\gamma\delta$ T1 lineage. **C.** Expression of genes defining DN subsets in merged datasets: *Cpa3* for DN2 and $\gamma\delta$ T cells, *Il2ra* (encodes CD25) for DN2/3 cells, *Ptcr* (encodes pre-Ta) for DN3 cells, and *Id3* for $\gamma\delta$ T cells and DN4 cells. *Cd8b1* is upregulated transcriptionally before surface expression and marks $\alpha\beta$ -T lineage commitment within DN4 cells. *Cd4* was undetectable, validating our MACS enrichment strategy. **D.** Expression of *Rorc* in WT versus Id3-KO cells in $\gamma\delta$ T cell subsets. **E.** *Rorc* expression in all WT and Id3-KO clusters.

Figure 6 - figure supplement 2. $\gamma\delta$ TCR⁺ cells from E18 Id3-KO mice include a population of CD4⁺CD8⁺ cells, indicating diversion to the $\alpha\beta$ -T lineage program. E18 fetal thymocytes were subjected to flow cytometry. Cells were gated on the TCR $\gamma\delta$ ⁺CD3⁺ population and analyzed for expression of CD4 and CD8 which was quantified in a bar graph depicting the percentage of CD4⁺CD8⁺ (DP) cells within the $\gamma\delta$ TCR⁺ population.

Figure 7 - figure supplement 1. Accessibility and occupancy of *Id3* locus elements by HEB, E2A, and Egr2 before and after pre-TCR or $\gamma\delta$ TCR signaling. **A.** *Rag2*^{-/-} DN3 cells were transduced (DN3 + $\gamma\delta$ TCR) or not (DN3) and stimulated with $\gamma\delta$ TCR ligand, followed by ChIP-seq analysis of HEB (green) and E2A (blue) binding. Peaks in the *Id3* locus were aligned to Egr2 ChIP data to locate overlapping sites of binding. These peaks were also aligned to context-specific accessibility peaks obtained from ATAC-seq analysis of *Rag2*^{-/-} DN3 (pre-selection) cells, and DN3 and DN4 (post-selection) cells from WT and HEB cKO mice. Sites showing overlapping HEB/E2A/Egr2 peaks and decreased accessibility in HEB cKO samples are designated as *HE1* and *HE2*. Circled peaks indicate a lower degree of accessibility in HEB cKO mice. **B.** Transcription factor binding motifs for HEB/E2A (E box; blue) and Egr2 (Egr; red) were identified in *HE1* and *HE2* in close proximity to each other. Colors of sequence bars indicate intensity of ChIP-seq signal, as shown in (A). It should be noted that HEB/E2A binding, but not Egr2 binding, is dampened in post-selection cells, as expected due to the increase in *Id3* expression. Ranges of ATAC-seq traces were kept constant to allow direct comparisons.

Figure 7 - figure supplement 2. Model for HEB and *Id3* requirements in the development and maturation of $\gamma\delta$ T17 cells. **A.** Strong $\gamma\delta$ TCR signaling induces high levels of Egr2, which are sufficient to drive *Id3* upregulation without HEB, whereas HEB is also required under lower $\gamma\delta$ TCR signaling conditions. Distinct cytokine signals also participate in *Id3* modulation and $\gamma\delta$ T cell lineage choice. **B.** $\gamma\delta$ T17 development occurs in two stages, the first of which is HEB-dependent, and the second of which is *Id3*-dependent. HEB induces *Id3* during the first stage, which acts in a negative feedback loop to inhibit HEB activity during the second stage. The absence of *Id3* allows higher E protein activity, which inhibits second stage regulators but also results in *Id2* upregulation, providing partial compensation for the loss of *Id3*.

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Peer reviews

Reviewer #1 (Public review):

The authors use Flow cytometry and scRNA seq to identify and characterize the defect in gdT17 cell development from HEB f/f, Vav-icre (HEB cKO) and Id3 germline-deficient mice. HEB cKO mice showed defects in the gdT17 program at an early stage, and failed to properly upregulate expression of Id3 along with other genes downstream of TCR signaling. Id3KO mice showed a later defect in maturation. The results together indicate HEB and Id3 act sequentially during gdT17 development. The authors further showed that HEB and TCR signaling synergize to upregulate Id3 expression in the Scid-adh DN3-like T cell line. Analysis of previously published Chip-seq data revealed binding of HEB (and Egr2) at overlapping regulatory regions near Id3 in DN3 cells. The study provides insight into mechanisms by which HEB and Id3 act to mediate gdT17 specification and maturation. The work is well performed and clearly presented.

Comments on revisions:

The authors have answered all of my questions. I am strongly supportive of the revised work. <https://doi.org/10.7554/eLife.109197.2.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript by Selvaratnam et al. defines how the transcription factor HEB integrates with TCR signaling to regulate Id3 expression in the context of gdT17 maturation in the fetal thymus. Using conditional HEB ablation driven by Vav Cre, flow cytometry, scRNA-seq, and reanalysis of ChIP-seq data the authors, provide evidence for a sequential model in which HEB and TCR-induced Egr2 cooperatively upregulate Id3, enabling gdT17 maturation and limiting diversion to the ab lineages. The work provides an important mechanistic insight into how the E/ID-protein axis coordinates gd T cell specification and effector maturation.

Strengths include:

- (1) The proposed model that HEB primes, TCR induces, and Id3 stabilizes gdT17 cells in embryonal development is elegant and consistent with the findings.
- (2) The choice of animal models and the study of a precise developmental window.
- (3) The cross-validation of flow, scRNA-seq, and ChIP-seq reanalyses strengthens the conclusions.
- (4) The study clarifies the dual role of Id3, first as an HEB-dependent maturation factor for gdT17 cells, and as a suppressor of diversion to the ab lineages.

Comments on revisions:

In this revised version of their manuscript the authors have effectively addressed all of my previous concerns. In its current form the study represents a significant advancement in our understanding of how the transcription factor HEB integrates with TCR signaling to regulate Id3 expression in the context of gdT17 maturation in the fetal thymus. In this revised version of their manuscript the authors have effectively addressed all of my previous concerns. In its current form the study represents a significant advancement in our understanding of how the transcription factor HEB integrates with TCR signaling to regulate Id3 expression in the context of gdT17 maturation in the fetal thymus.

<https://doi.org/10.7554/eLife.109197.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors of this manuscript have addressed a key concept in T cell development: how early thymus gd T cells subsets are specified and the elements that govern gd T17 versus other gd T cell subset or ab T cell subsets are specified. They show that the transcriptional regulator HEB/Tcf12 plays a critical role in specifying the gd T17 lineage and, intriguingly that it up regulates the inhibitor Id3 which is later required for further gd T17 maturation.

Strengths:

The conclusions drawn by the authors are amply supported by a detailed analysis of various stages of T cell maturation in WT and KO mouse strains at the single cell level both phenotypically, by flow cytometry for various diagnostic surface markers, and transcriptionally, by single cell sequencing. Their conclusions are balanced and well supported by the data and citations of previous literature.

Weaknesses:

I actually found this work to be quite comprehensive.

Comments on revisions:

Nothing to add here. The authors were very thorough in their original submission, and all minor issues identified have been addressed to my satisfaction.

<https://doi.org/10.7554/eLife.109197.2.sa1>

Author response:

The following is the authors' response to the original reviews.

We thank the reviewers for their enthusiasm and insightful suggestions. Our responses to specific concerns and questions are detailed below.

Public Reviews:

Reviewer #1 (Public review):

The authors use Flow cytometry and scRNA seq to identify and characterize the defect in gdT17 cell development from HEB f/f, Vav-icre (HEB cKO), and Id3 germline-deficient mice. HEB cKO mice showed defects in the gdT17 program at an early stage, and failed to properly upregulate expression of Id3 along with other genes downstream of TCR signaling. Id3KO mice showed a later defect in maturation. The results together indicate HEB and Id3 act sequentially during gdT17 development. The authors further showed that HEB and TCR signaling synergize to upregulate Id3 expression in the Scid-adh DN3-like T cell line. Analysis of previously published ChIP-seq data revealed binding of HEB (and Egr2) at overlapping regulatory regions near Id3 in DN3 cells.

The study provides insight into mechanisms by which HEB and Id3 act to mediate gdT17 specification and maturation. The work is well performed and clearly presented. We only have minor comments.

Reviewer #2 (Public review):

Summary:

The manuscript by Selvaratnam et al. defines how the transcription factor HEB integrates with TCR signaling to regulate Id3 expression in the context of gdT17 maturation in the fetal thymus. Using conditional HEB ablation driven by Vav Cre, flow cytometry, scRNA-seq, and reanalysis of ChIP-seq data the authors, provide evidence for a sequential model in which HEB and TCR-induced Egr2 cooperatively upregulate Id3, enabling gdT17 maturation and limiting diversion to the ab lineages. The work provides an important mechanistic insight into how the E/ID-protein axis coordinates gd T cell specification and effector maturation.

Strengths include:

(1) The proposed model that HEB primes, TCR induces, and Id3 stabilizes gdT17 cells in embryonal development is elegant and consistent with the findings.

(2) The choice of animal models and the study of a precise developmental window.

(3) The cross-validation of flow, scRNA-seq, and ChIP-seq reanalyses strengthens the conclusions.

(4) The study clarifies the dual role of Id3, first as an HEB-dependent maturation factor for gdT17 cells, and as a suppressor of diversion to the ab lineages.

Weaknesses:

(1) The ChIP-seq reanalysis indicates overlapping HEB, E2A, and Egr2 peaks ~60 kb upstream of Id3. Given that the Egr2 data are not generated using the same thymocyte subsets, some form of validation should be considered for the co-binding of HEB and Egr2, potentially ChIP-qPCR in sorted gdT17 progenitors.

We agree that this is a valid concern and continue to work on confirming the mechanism from several other angles. Validating HEB/E2A and Egr2 co-binding in gdT17 cell progenitors by ChIP-qPCR would/will be a very precise and definitive experiment, but it will be very challenging to perform, in part due to the low numbers of gdT17 precursors in the fetal thymus (note the y-axis scales in Fig. 1F, J). As a complementary approach, we have analyzed additional ChIP-seq data for HEB/E2A binding in Rag2^{-/-} DN3 cells retrovirally transduced with the KN6 gdTCR cultured with stroma expressing the weak KN6 ligand T10 for 4 days. This analysis revealed that the binding of HEB/E2A on those sites persisted after weak gdTCR signaling, strengthening the likelihood that concurrent binding of HEB/E2A and Egr2 occurs during this developmental transition. We noted that HEB/E2A binding was slightly dampened in Rag2^{-/-} DN3 + gdTCR cells relative to Rag2^{-/-} DN3 cells, consistent with the induction of Id3 and subsequent Id3-mediated disruption of E protein binding. We also located HEB/E2A and Egr binding sites in close proximity in the two regions that shared peaks between HEB/E2A and Egr2 analyses (HE1 and HE2), in line with the potential participation of these two transcription factors in an enhanceosome binding complex.

Furthermore, we examined the chromatin landscape of the Id3 locus by sorting WT DN3 and DN4 cells, as well as Rag2^{-/-} DN3 cells to provide a genuine pre-selection context, and performing ATAC-seq (Figure 7–suppl 7A). Given the known ability of E2A and HEB to induce chromatin remodeling, we also examined accessibility in DN3 and DN4 cells from HEB cKO mice. Alignment of ATAC-seq and ChIP-seq peaks in the Id3 locus revealed accessibility of HE1 and HE2 in Rag2^{-/-}, WT DN3, and WT DN4 cells. However, accessibility of HE1 and HE2 was dampened in HEB cKO cells, especially at the DN3 stage, suggesting that HEB may be involved in remodeling the Id3 locus, resulting in a poised state that enables TCR-dependent transcription factors to induce Id3 proportionally to TCR signal strength. These data are now presented as a new “Figure 7 – figure supplement 1” with corresponding Results, Discussion, and Methods updates.

Our next story will be focused on a finer dissection of the Id3 cis-regulatory elements and their combinatorial regulation by HEB/E2A and other transcription factors, and how they relate to specific signaling pathways. For this study, we will modify the language regarding Egr2 to reflect the open questions that still remain to be addressed.

(2) E2A expression is not affected in HEB-deficient cells, raising the question of partial compensation, a point that should be specifically discussed.

This confounding factor is always an issue with E proteins. We have now added a section to the discussion that highlights previous literature and relates it to our findings.

(3) All experiments are done at E18, when fetal gdT17 development predominates. The discussion could address whether these mechanisms extend to neonatal or adult gdT17 subsets.

In our 2017 paper (PMID 29222418) we showed that HEB cKO mice have defects in the production of functional gdT17 cells in fetal and neonatal thymus and in the adult periphery (in lungs and spleen). While the adult thymus does not support the development of fully

functional innate gd T cells, it does contain gdTCR⁺ cells that have activated the Sox-Maf-Rorc network (Yang 2023, PMID 37815917). It will be very interesting to assess the impact of HEB loss on these cells, and we are actively pursuing this goal. For now, we will add a paragraph to the discussion addressing what we know from previous work and what is yet to be learned.

Reviewer #3 (Public review):

Summary:

The authors of this manuscript have addressed a key concept in T cell development: how early thymus gd T cell subsets are specified and the elements that govern gd T17 versus other gd T cell subsets or ab T cell subsets are specified. They show that the transcriptional regulator HEB/Tcf12 plays a critical role in specifying the gd T17 lineage and, intriguingly, that it upregulates the inhibitor Id3, which is later required for further gd T17 maturation.

Strengths:

The conclusions drawn by the authors are amply supported by a detailed analysis of various stages of T cell maturation in WT and KO mouse strains at the single cell level, both phenotypically, by flow cytometry for various diagnostic surface markers, and transcriptionally, by single cell sequencing. Their conclusions are balanced and well supported by the data and citations of previous literature.

Weaknesses:

I actually found this work to be quite comprehensive. I have a few suggestions for additional analyses the authors could explore that are unrelated to the predominant conclusions of the manuscript, but I failed to find major flaws in the current work.

I note that HEB is expressed in many hematopoietic lineages from the earliest progenitors and throughout T cell development. It is also noteworthy that abortive gamma and delta TCR rearrangements have been observed in early NK cells and ILCs, suggesting that, particularly in early thymic development, specification of these lineages may have lower fidelity. It might prove interesting to see whether their single-cell sequencing or flow data reveal changes in the frequency of these other T-cell-related lineages. Is it possible that HEB is playing a role not only in the fidelity of gdT17 cell specification, but also perhaps in the separation of T cells from NK cells and ILCs or the frequency of DN1, DN2, and DN3 cells? Perhaps their single-cell sequencing data or flow analyses could examine the frequency of these cells? That minor caveat aside, I find this to be an extremely exciting body of work.

Excellent question, and the underlying answer is yes, loss of HEB renders the cells more open to divergence to non-T lineages, even at the DN3 stage. Although our datasets did not reveal those cells, we have examined this question previously. In our 2011 paper (Braunstein, 2011, PMID 21189289) where we identified “DN1-like” cells arising from HEB^{-/-} DN3 cells in OP9-DL1 co-cultures. These cells responded to IL-15 and IL-7 by differentiating into cytotoxic NK-like cells. We did not detect TCRb rearrangements but did not look for gdTCR rearrangements. Subsequently, multiple papers from other labs showed that ILC2 were greatly expanded in the thymus using Id-overexpression transgenic mice and HEB/E2A-double deficient mice (Miyazaki, 2023, PMID 28514688; Miyazaki, 2025, PMID 39904558; Berrett, 2019, PMID 31852728; Qian, 2019, PMID 30898894; Peng, 2020, PMID:32817168). The ILCs in these mice had TCRg rearrangements, consistent with a shared origin with WT thymic-derived ILCs. In unpublished data from our lab, we found an increase in the numbers of ILC2 but not ILC3 in HEB cKO fetal thymic organ cultures. We did not follow up on this work any further since the

topic was being heavily pursued in other labs, but remain very interested in this branchpoint, and will mention the literature in the discussion.

Joint recommendations for the authors:

(1) Experimental validation (for mechanistic clarity)

The ChIP-seq reanalysis indicates overlapping HEB, E2A, and Egr2 peaks ~60 kb upstream of Id3. Given that the Egr2 data are not generated using the same thymocyte subsets, some form of validation should be considered for the co-binding of HEB and Egr2, potentially ChIP-qPCR in sorted gdT17 progenitors to substantiate the proposed cooperative mechanism.

See above; new experiments with ATAC-seq and additional ChIP-seq analysis.

(2) Figures

Potential inconsistencies in Figure 1H: In the legend to Figure 1H, Vg1-Vg5- cells are considered Vg6+ cells. Flow plots show reduced A Vg1-Vg5- population in HEBc ko mice, but the accompanying bar plot shows increased frequency of Vg6+ cells.

Vg6 cells are actually considered to be Vg4-Vg5-Vg1- cells (not Vg4- Vg1- cells, which is important in the fetal context). The flow plot shows the percentage of Vg6 cells out of the Vg1-Vg4- population, whereas the bar plot shows the percentage of Vg6 cells out of all gdTCR+ cells. The ratio of Vg6 to Vg5 cells decreases within the Vg1-Vg4- population, whereas the overall percentages and numbers of Vg6 cells in all gd T cells is increased in HEB cKO mice. We have now more clearly explained this in the text and the figure legend.

Clarify which cells produce IL-17A in Figure 1L.

This plot is gated on all gd T cells stimulated with PMA/ionomycin; this has been added to the results and figure legend.

In Supplementary Figure 2, legend, do the authors mean that TRGV4 was depleted? The authors write TRDV4. Please check.

Thank you for catching this mistake, we have corrected it.

In Figure 7, the Author showed Id3 mRNA expression. Can the expression of Id2 be included?

That is a really interesting question, and we will follow up on it in future studies.

If Id1 or Id4 are relevant for any of these studies, can their expression be shown in Supplementary Figure 3A? If these are minimally expressed or not expressed, this could be mentioned.

Id1 and Id4 were not detectable in our studies, this is now stated in the results section describing expression of E proteins and Id proteins.

(3) Discussion

Discuss possible redundancy between HEB and E2A, as E2A expression appears unaffected in HEB-deficient cells.

See above

Address whether the mechanisms identified at E18 (embryonic stage) also apply to neonatal or adult $\gamma\delta$ T17 subsets.

See above

Expand on how HEB function may relate to other hematopoietic or early lymphoid lineages (NK/ILC, DN1-DN3 stages), based on reviewer curiosity.

See above

(4) Methods and terminology

Define the terms $\gamma\delta Te1$ and $\gamma\delta Te2$ (e.g., early effector subsets).

This has been defined more clearly in several sections of the text.

Add details to the scRNA-seq methods section (average number of cells analyzed and sequencing depth per cell).

These details have been added.

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